

Smart energy strategies: Leveraging LSTM and LLMs for advanced energy management

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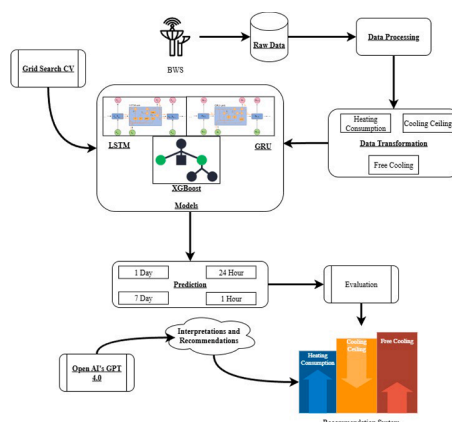
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HIGHLIGHTS

- Proposed a dynamic recommendation system for predicting and managing SH, CC, and FC energy use
- Applied LSTM, GRU, and XGBoost models to assess accuracy and efficiency in energy forecasting.
- Demonstrated that LSTM provides superior performance for sustainable building energy prediction.
- Enhanced energy efficiency and sustainability using ML to adjust building operations in real time.
- Integrated OpenAI's GPT-4 to interpret energy graphs and deliver context-specific recommendations.

GRAPHICAL ABSTRACT



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ABSTRACT

Accurately predicting and optimizing heating and cooling demands in building energy management is crucial for enhancing efficiency and reducing energy consumption. Traditional methods often struggle with building energy usage patterns' nonlinear and variable nature. With the advent of advanced data collection through smart sensors, there is a growing need for intelligent systems to leverage this data to provide actionable insights. This study addresses the gap by developing a recommendation system using three machine and deep learning models, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and XGBoost to predict and optimize the efficiency levels of space heating, ceiling cooling, and free cooling systems. Our proposed solution harnesses the power of these models, with LSTM performing best overall, to forecast energy consumption across hourly and daily timescales, enabling precise adjustments and efficient energy management. The methodology involves extensive data preprocessing, including hierarchical imputation of missing values and label encoding of categorical variables, followed by the transformation of raw data into efficiency levels. The deep learning model architecture, consisting of sequential layers, captures long-term dependencies in the data, while grid search-based hyperparameter tuning optimizes model performance. Results indicate high predictive accuracy, with R-

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squared values demonstrating the model's ability to explain up to 97.2 % of the variance in hourly space heating, 95.2 % in daily ceiling cooling, and 93 % in daily free cooling energy consumption. Additionally, we interpret graphs using OpenAI's GPT-4 model to enhance understanding and facilitate actionable insights. This interpretation enhances the clarity of the predictive results, supporting more informed decision-making in energy management. The significance of this work lies in its potential to transform energy management practices in building environments, providing a robust tool for optimizing heating and cooling operations and contributing to overall energy efficiency.

1. Introduction

Buildings account for a significant portion of global primary energy use (40 %) and greenhouse gas emissions (30 %) [1]. Enhancing energy efficiency in buildings, crucial for mitigating environmental impact, focuses on critical energy consumers like lighting, cooling, and heating [2–4]. Understanding dynamic thermal behaviors under non-stationary conditions is essential [5]. ASHRAE emphasizes energy source understanding for optimizing building operations, noting space conditioning, lighting, and plug loads as major energy end-uses [6]. Passive (e.g., double skin façade) and active (e.g., variable refrigerant flow systems) technologies offer avenues for improving energy efficiency [7]. Simulation studies show potential savings and performance improvements in building systems [8,9].

Thus, to reduce energy consumption, energy modeling should be incorporated into the early design stage, and further building usage should be used to predict energy consumption in various and unpredictable circumstances [6]. Forecasting demands that depend on various building factors with nonlinear temporal characteristics is not easy; however, due to intelligent sensors' development and big data technologies, the application of data-driven models for thermal load and energy forecasting in buildings has improved [10]. These models have been receiving attention because they are flexible, simple, and easy to use; various algorithms are available to assess and forecast the energy requirements of buildings [11].

Several approaches have been considered in exploring machine learning and deep learning models applicable to energy consumption prediction in recommendation system. Standard methods such as linear regression and decision trees are easily explainable. Still, they can perform poorly in datasets with high temporal dependency non-linearity in the building energy data [12,13]. Other models, like random forests and gradient boosting machines, can handle non-linear relationships but may require extensive feature engineering and lack the ability to model sequential data effectively. Convolutional neural networks (CNNs) have shown promise in spatial data analysis but may not capture long-term dependencies well [14]. Nonetheless, deep learning models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures have shown effectiveness in modeling sequential data, particularly in energy consumption tasks [15].

This study develops an LSTM-based model and compares its performance with GRU and XGBoost, establishing a comparative framework to identify the most accurate approach for optimizing energy efficiency in buildings. LSTMs can learn long-term dependencies in data and transition between short and long-term dependencies, which are essential in predicting energy consumption as influenced by the environmental conditions and occupancy patterns in buildings [16].

Our research focused on developing a recommendation system for optimizing building energy efficiency, and three critical types of energy consumption will be targeted: space heating (SH), cooling ceiling (CC), and free cooling (FC). Space Heating stands for the energy required to warm the interiors during cold months and elevate temperatures ideal for users [17]. Cooling Ceiling involves the energy expended in cooling indoor spaces, particularly during warmer months, using ceiling-based cooling systems to maintain comfortable indoor temperatures [18]. On the other hand, Free Cooling requires an external air or water source to cool the internal environment without mechanical refrigeration and

relies on environmentally friendly conditions to conserve energy [19].

Sustainability in building operations is increasingly critical amid global efforts to mitigate climate change and reduce environmental impact [20,21]. Hence, our research aims to support sustainable objectives using effective and efficient energy predictors such as the LSTM model. Thus, besides presenting an opportunity to minimize energy waste, the proposed recommendation system for SH, CC, and FC is designed to predict and manage energy use dynamically.

One of the most critical aspects of this study is the interpretation of graphical analyses using OpenAI's Generative Pretrained 4.0 (GPT-4.0) model, which presents significant challenges and opportunities. One major challenge is accurately conveying complex patterns in energy consumption data, which can be influenced by various factors such as seasonal changes, occupancy patterns, and weather conditions. The graphical representations of space heating, ceiling cooling, and free cooling systems often reveal intricate interactions that are difficult to interpret without advanced tools. The integration of the GPT-4 model enhances this process by providing detailed, context-specific insights and recommendations based on the visualized data. This approach simplifies understanding of complex graphs and facilitates actionable decision-making by highlighting key trends and anomalies. To the best of our knowledge, this is the first recommendation system that integrates such interpretation capabilities, offering a more nuanced and precise understanding of energy consumption patterns. The significance of this interpretation lies in its ability to support more effective and informed energy management strategies, leading to improved efficiency and reduced operational costs in building environments. The contributions of the study are as follows:

- i. Development of a novel recommendation system using LSTM model to predict and optimize SH, CC, and FC energy consumption in buildings.
- ii. Comparative evaluation of LSTM against GRU and XGBoost models to identify the most effective architecture for forecasting building energy consumption.
- iii. Enhancement of energy efficiency and sustainability by leveraging advanced machine learning techniques to dynamically adjust building energy operations based on real-time environmental factors and occupancy patterns.
- iv. Introduced a novel recommendation system that integrates OpenAI's GPT-4 for interpreting graphical analyses of energy consumption, providing detailed, context-specific insights to enhance decision-making and operational efficiency.

2. Related work

Recommendations for harmonizing heating and cooling systems have been shown to improve building energy efficiency [22]. Prior works have leveraged data analytics and machine learning to generate tailored HVAC optimization strategies based on historical consumption, environmental conditions, and user preferences [23], typically suggesting setpoint adjustments, schedule optimization, and other operational refinements to balance comfort and efficiency.

Reinforcement learning (RL) approaches have further advanced these systems by enabling adaptive, context-aware control strategies. For example, recEnergyW engages occupants in energy-saving practices

using Markov decision processes [24], while other studies have applied deep reinforcement learning (DRL) to dynamically adjust indoor temperature setpoints for higher photovoltaic (PV) utilization, achieving measurable savings and comfort improvements [25]. Additional RL-based frameworks, such as HERS [26], integrate human preferences for demand-side management, though these often exclude PV, storage, or microgrid contexts. Comprehensive RL-based HVAC optimization frameworks have also been proposed [27], but real-world scalability and computational cost remain ongoing challenges.

Traditional recommendation methods, including model-based expert systems and rule-based analytics, have been deployed in commercial and residential settings [28,29], often using real-time building data to detect inefficiencies and propose behavioral adjustments. While these methods can be effective, they typically require extensive manual configuration, have limited adaptability to diverse building types, and lack integration with advanced learning-based models.

Recent studies have incorporated deep learning into recommendation systems, leveraging IoT and edge computing for scalable, real-time analysis [30]. However, many still rely on static, rule-based suggestions and do not fully exploit collaborative filtering or goal-oriented personalization. LSTM-based models have also been applied to building energy forecasting, with enhancements through transfer learning to improve generalization across weather conditions [28]. Yet, these works have not addressed automated interpretation of predictive outputs, and their ability to generalize across diverse building types remains limited.

While LSTM and other forecasting models have been extensively applied for energy prediction, there is a lack of systems that automatically translate forecasted results into actionable recommendations without human intervention. To our knowledge, this is the first work to integrate OpenAI's GPT-4 into an LSTM-based building energy management pipeline. GPT-4's natural language interpretation capabilities enable the transformation of multi-dimensional prediction outputs into context-specific operational strategies, bridging the gap between raw forecasts and decision-making. This integration addresses a limitation in prior studies where interpretation and recommendation remained either manual or rule-based. The novelty lies in combining LSTM forecasting with GPT-4-driven interpretation for real-time, scalable, and building-specific energy optimization (Table 1).

3. Dataset

The weather variables dataset used for our recommendation system for cooling, free cooling, and heating consumption is sourced from a specialized weather station atop the European Central Bank (ECB) skyscraper. These data sets offer beneficial climatic conditions, including air temperature, relative humidity, wind speed, and other factors related to the building's surrounding environment. The height of the position is maintained at such a level to capture data that very well correlate with the existing atmosphere of the building.

Appending with Building Automation System (BAS) augments the access and use of this data. Due to the integrated nature of the BAS's data transmission and storage, control rooms can provide building managers and operators with current meteorological data. This capability supports current decision-making processes based on current weather conditions.

Also, it identifies other benefits of BAS's ability to regulate HVAC systems. This way, by incorporating trends and patterns, consumers and other stakeholders standing from the building can see how specific weather parameters influence cooling consumption, free cooling consumption, and heating consumption over time. This statistic accumulation is beneficial for detailed analysis to accentuate relationships between the weather variables and the measures of energy usage.

3.1. Space heating consumption and air temperature

The spatial distribution of the space heating consumption and the external air temperature is well illustrated in Fig. 1, collected by the Building Weather Station (BWS). The results show a strong negative relationship, as observed from the assembled data point. As air temperature declines, there is a notable increase in heating consumption, reaching its zenith within the temperature range of -10°C to 0°C . However, as the temperatures go beyond 0 and above, there is a noted decrease in heating consumption, gradually dropping to a negligible level as the external temperature nears 15°C . Focusing on the density points under 15°C indicates a high level of heating demand under lower temperatures, suggesting that external temperature control is subject to the building's energy plan.

Table 1
Comparison of existing recommendation systems.

Ref.	Year	Focus	Dataset	Technique	Result	Limitation
[24]	2020	Energy-saving recommender system	NA	Deep reinforcement learning	energy reduction from 19 % to 26 %	Manual post-filtering due to limited recommendation acceptance data
[25]	2021	Smart home energy management	Data was gathered at 30 min intervals for each day over three months.	Deep reinforcement learning (DRL)	Average of 8 % energy savings, up to 16 % savings over summer	Challenges in obtaining user preferences and adapting to changes in PV production policies
[26]	2022	Smart home energy optimization	ARAS Activity Dataset	Deep reinforcement learning	Improved electricity cost optimization and comfort	Excludes devices with PV sources, energy storage, microgrid, or data sharing
[27]	2020	HVAC control and demand response optimization	TMY3	Reinforcement learning	Up to 22 % weekly energy reduction	• Challenges in real-world deployment and computational efficiency Top of Form
[28]	2022	Energy-saving recommender system	Ten pilot buildings	Model-based expert system	overall energy saving achieved is 12.2 %	Manual configuration and limited metadata
[29]	2018	Occupant-informed energy-saving recommendations	Historical location data	Gaussian fit	25 % more energy savings	Reliance on historical location data
[22]	2020	recommendations for human engagement	Real-time sensor data	Recommendation engine leveraging sensor and actuator data	26 % increase on the recommendation acceptance ratio	scalability and user acceptance beyond initial findings
[30]	2023	Energy efficiency through IoT	5 different offices	LSTM	accuracy of 89.06 %	Further development is needed for goal-based and collaborative filtering scenarios.
[31]	2023	Building energy load forecasting	Hospital building prototype from the U.S. DOE	LSTM, Transfer Learning	R-sq: 0.97	Practical application concerns across diverse building types and weather conditions

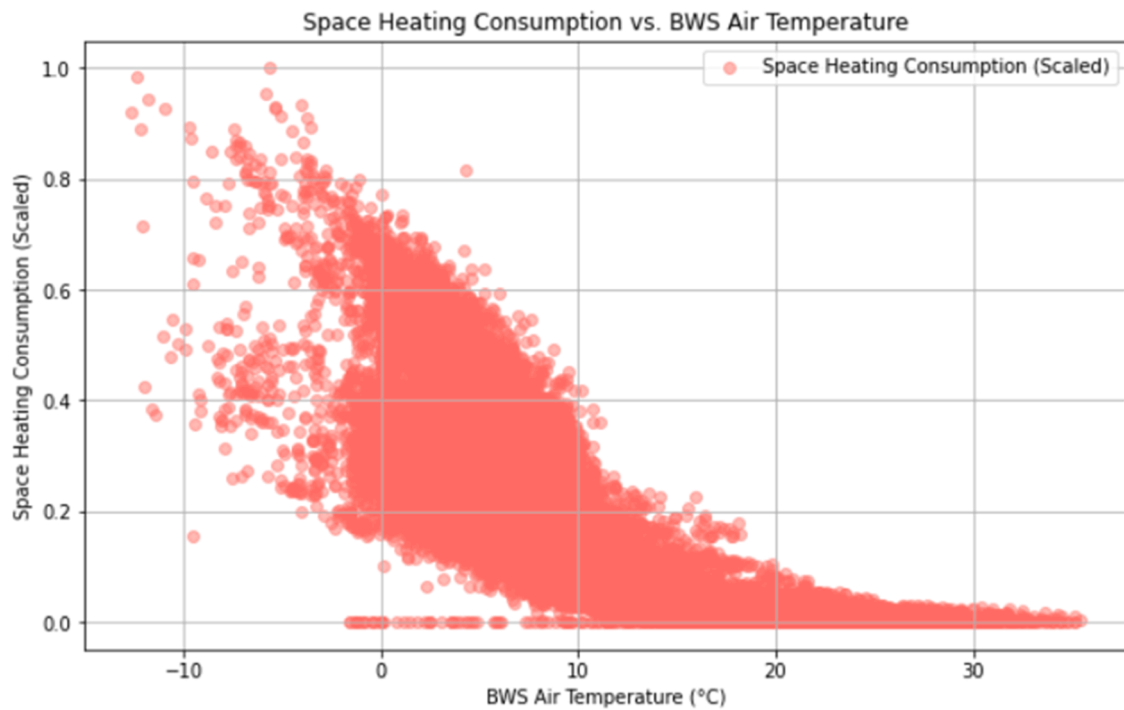


Fig. 1. Relationship between space heating consumption and external air temperature.

3.2. Ceiling cooling and air temperature

As can be seen in Fig. 2, ceiling cooling consumptions correlate with BWS air temperatures. The plot demonstrates a clear trend: cooling consumption remains negligible until approximately 7 °C. Above this threshold, there is a rapid increase in cooling consumption as the air temperature rises, with the highest densities of data points observed between 15 °C and 30 °C. The decline in this trend shows that cooling systems are mostly initiated at higher temperatures to achieve

comfortable indoor temperatures. The extremely great spread of data points, precisely above 7 °C, emphasizes that cooling efficiency at higher temperatures is much higher and highlights the need for efficient indoor cooling. This means that most cooling load occurs in a specific temperature interval, indicating that the building's operation responds to the outside temperature.

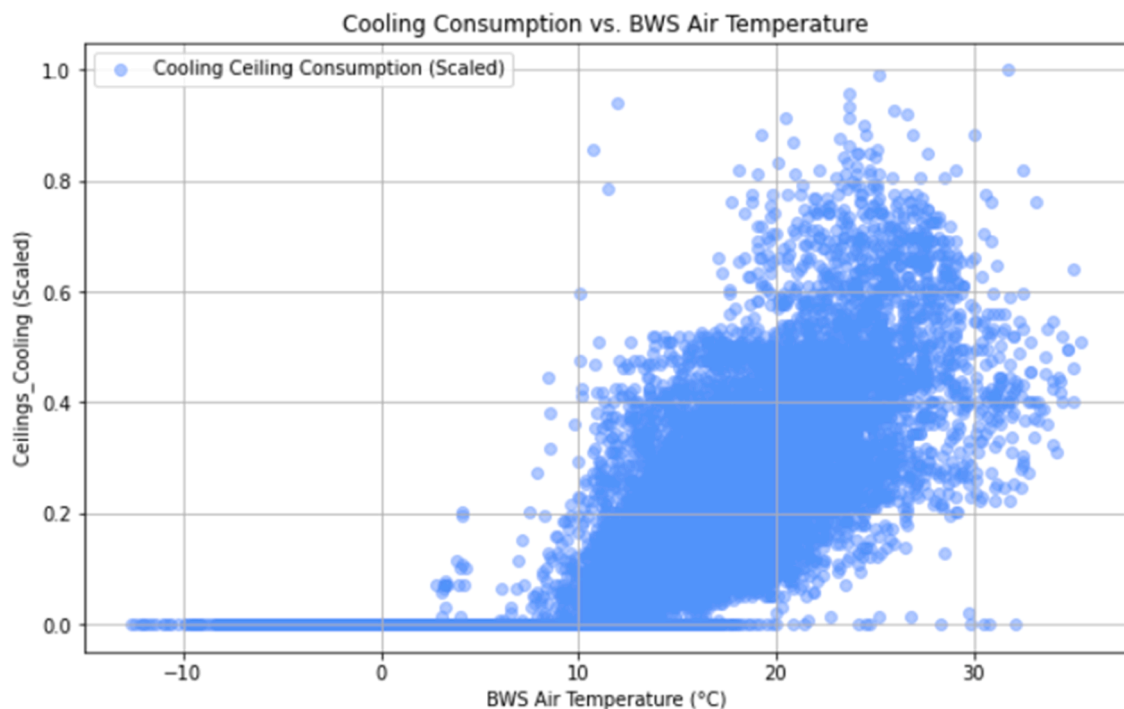


Fig. 2. Relationship between cooling consumption and external air temperature.

3.3. Free cooling and air temperature

Fig. 3 plots free cooling consumption with the air temperature in the building with BWS. The plot describes a quite different scenario where free cooling is used mainly when the external temperature is between 0 °C and 15 °C. This range is essential, as can be seen at 15 °C, above which the utilization of free cooling is considerably low. This range signifies that the system has the highest data point density when running within its free cooling operational zone, which uses cooler outside air to minimize the need for mechanical cooling. Most notably, the relative density of data points is exceptionally high within the approximate range of 0–10 °C, which seems to coincide with the peak of free cooling episodes. When the outside temperature gets below 0 °C or over 15 °C, the use of free cooling reduces, implying that the efficiency is optimum at moderate external temperatures; it demonstrates that free cooling methods should be incorporated into power utilization strategies, especially where the external weather is within the optimum operating range, as shown here most of the time.

3.4. Interaction between three energy consumption

Fig. 4 illustrates the scaled consumption of three types of energy, space heating, ceiling cooling, and free cooling, across winter, spring, summer, and fall from January 2020 to October 2022. Throughout the observed period, a clear seasonal pattern is evident. Space heating consumption peaks during winter, reflecting increased heating requirements to maintain indoor comfort in colder conditions. Conversely, ceiling cooling consumption surges during the summer months, aligning with higher external temperatures and the subsequent need for cooling. Free cooling displays variability across all seasons but is most prominent during transitional seasons, such as spring and fall, where mild external temperatures facilitate the utilization of outdoor air for cooling purposes. Notably, the overlapping periods of high consumption for heating and cooling suggest opportunities for further optimization, particularly in managing peak loads and enhancing the overall energy efficiency of the building systems.

4. Proposed methodology

This study focuses on developing a Long Short-Term Memory (LSTM) neural network model to predict the efficiency levels of ceiling cooling, free cooling, and heating consumption, with the goal of supporting an energy recommendation system. The LSTM model is evaluated against two established baselines: Gated Recurrent Unit (GRU) and XGBoost, allowing a comparative analysis of their effectiveness in both daily and hourly prediction tasks. These comparisons help assess the suitability of each model for optimizing energy management in building environments. Fig. 5 provides a comprehensive overview of the proposed approach.

4.1. Data preparation and transformation

Our analysis utilizes historical meteorological and energy consumption data from the ECB skyscraper's Building Weather Station (BWS). This data set contains influential parameters like air temperature, humidity, wind speed, and energy used for ceiling cooling, free cooling, and heating.

The preprocessing starts by selecting the numeric ones other than 'Year,' 'Month,' and 'Hour' as the non-numeric columns. The missing values are then imputed hierarchically by group means, and 'Year,' 'Month,' and 'Hour' is the priority. The group replaces further imputations of missing values, meaning values obtained by 'Year' and 'Month,' then 'Year' and 'Season.' Last of all, if some missing values remain in each variable, they are replaced by the annual average. Some variables are converted from categorical to numerical by applying the label encoding on 'Season' and 'Weekday.'

To improve the assessment and readability of models, raw data is first converted to efficiency levels between 0 and 100 %. This transformation facilitates standardization of the intensity of energy-consuming systems, thus helping the model calculate the relativity of energy utilization using normative season and operating conditions.

The given dataset is divided into training, validation, and testing data subsets to train and validate the model. The temporal features are arranged in sequences to take account of temporal dependencies in the

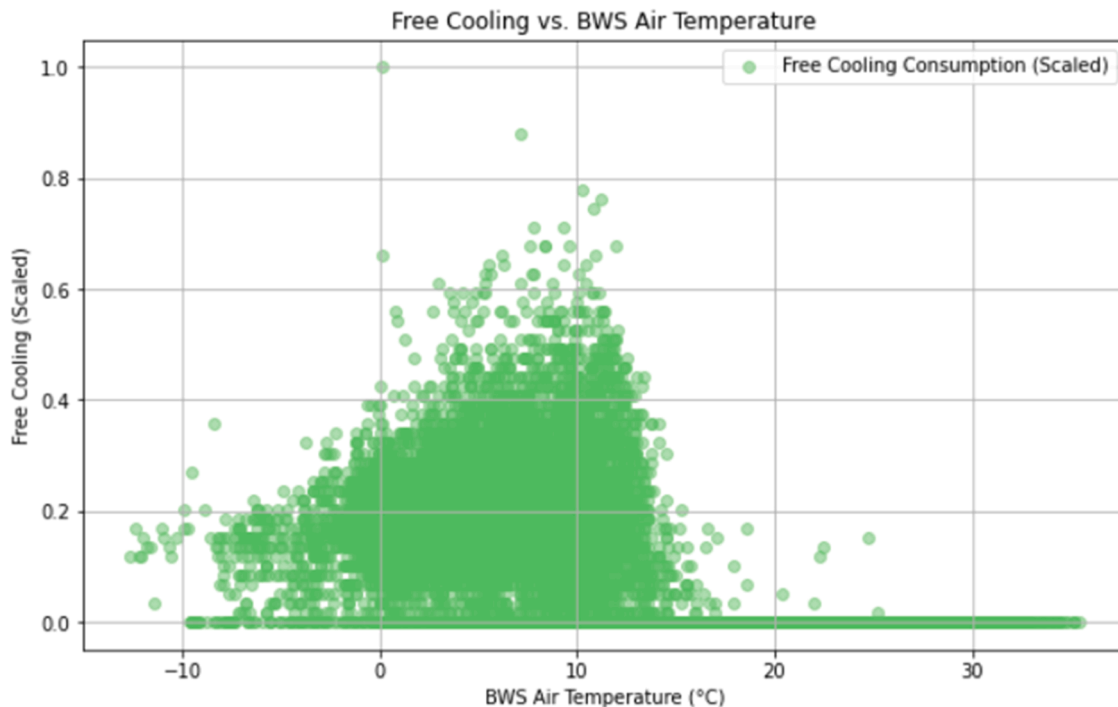


Fig. 3. Relationship between free cooling and external air temperature.

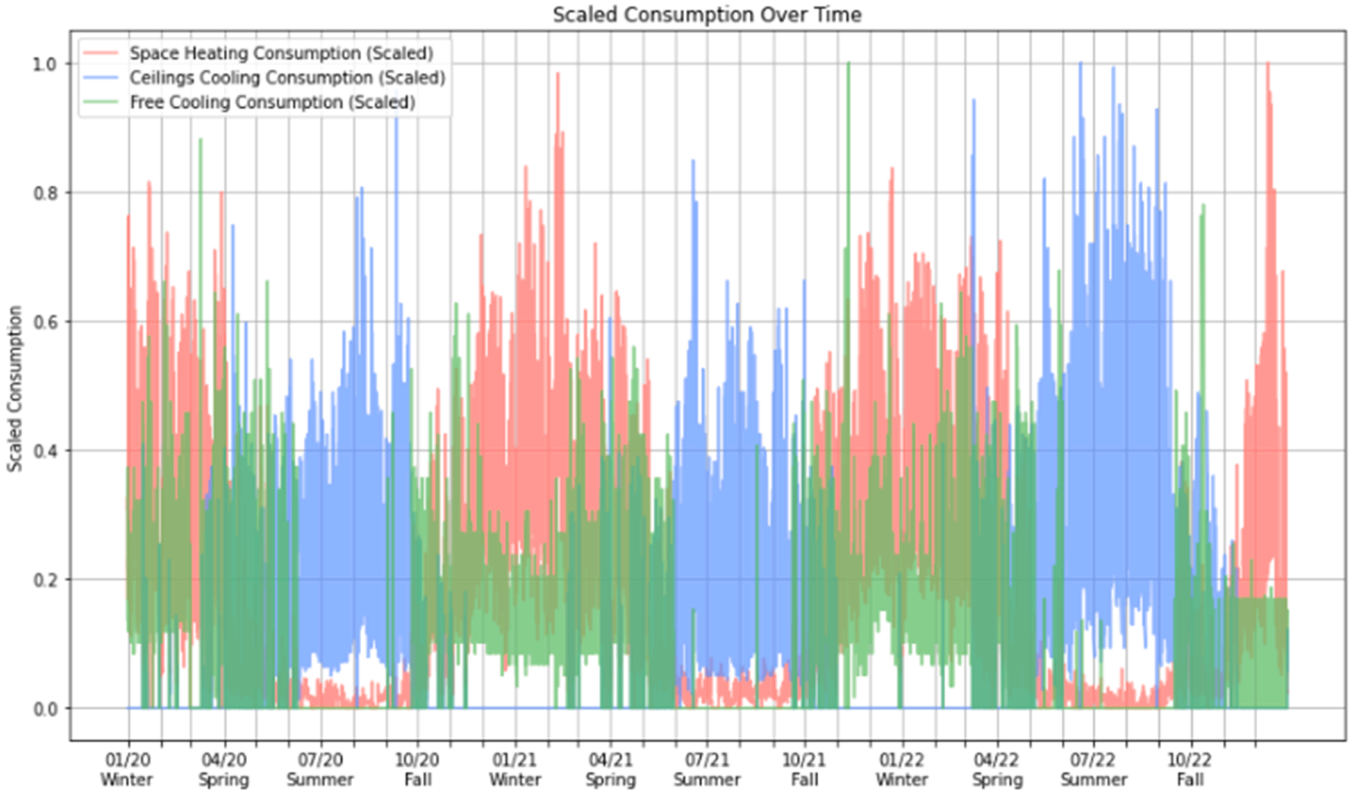


Fig. 4. Scaled consumption of space heating, ceiling cooling, and free cooling over time across different seasons.

given energy usage patterns.

4.2. Model architecture

We adopt a sequential LSTM model due to its demonstrated effectiveness in modeling time series data with long-term dependencies. The architecture consists of LSTM layers followed by a dense output layer. Each LSTM layer uses the ‘tanh’ activation function, enabling stable gradient flow and the capture of non-linear temporal relationships across energy consumption sequences. The dense output layer uses ‘ReLU’ activation to produce three output values corresponding to the predicted efficiencies of ceiling cooling (CC), free cooling (FC), and space heating (SH) consumption. In the proposed LSTM architecture, the model complexity and overfitting risks were addressed through the addition of an extra LSTM layer, which improved the model’s capacity to capture temporal dependencies while maintaining stability during training. Furthermore, cross-validation was used to evaluate model performance across different data partitions, ensuring robustness and generalizability within the target building dataset. This approach provided a balance between model expressiveness and control over overfitting without relying on dropout-based regularization or early stopping mechanisms.

In addition to the LSTM, a comparative model based on GRU (Gated Recurrent Unit) is developed with an identical architecture for performance benchmarking. GRU, while similar to LSTM, simplifies the gating mechanism and is evaluated to understand trade-offs between computational efficiency and accuracy. Both LSTM and GRU models are tuned using a systematic hyperparameter search.

For baseline and comparative purposes, an XGBoost model is also implemented. XGBoost does not model temporal dependencies directly but is included for its strong performance in tabular regression tasks. The XGBoost model is adapted for multi-output prediction and serves as a reference for evaluating the advantages of deep sequence-based models. The architecture of the proposed LSTM and GRU model is summarized in

Table 2.

4.3. Training and hyperparameter tuning

A rigorous training and hyperparameter optimization process is followed for all three models (LSTM, GRU, and XGBoost). For the sequence models, time series cross-validation (3-fold) is employed using GridSearchCV to ensure model generalization across different temporal segments. The LSTM and GRU models are each trained for four different forecasting scenarios:

1. **Daily LSTM/GRU (1-day)** – Predicts energy efficiency for the next day using daily-aggregated data.
2. **Daily LSTM/GRU (7-day)** – Forecasts energy usage over the next 7 days to support weekly energy planning.
3. **Hourly LSTM/GRU (1 h)** – Provides high-resolution predictions for immediate adjustments.
4. **Hourly LSTM/GRU (24 h)** – Supports day-ahead scheduling through hourly forecasts for the next 24 h.

The hyperparameter grid for LSTM and GRU models includes the number of units, activation function, optimizer type, and learning rate. The best configuration was found to be 150 units, ‘relu’ activation, ‘adam’ optimizer, a learning rate of 0.001, with 20 epochs and a batch size of 16. The optimal hyperparameters, which yielded the best performance, are detailed in Table 3.

For the XGBoost model, RandomizedSearchCV is employed due to the high-dimensional parameter space and computational constraints (Table 4). The model is evaluated using 3-fold cross-validation, and the best hyperparameters are selected based on the lowest negative mean squared error. The search space includes core parameters such as number of estimators, learning rate, maximum tree depth, and regularization terms.

For all models, the objective is to minimize the Mean Squared Error

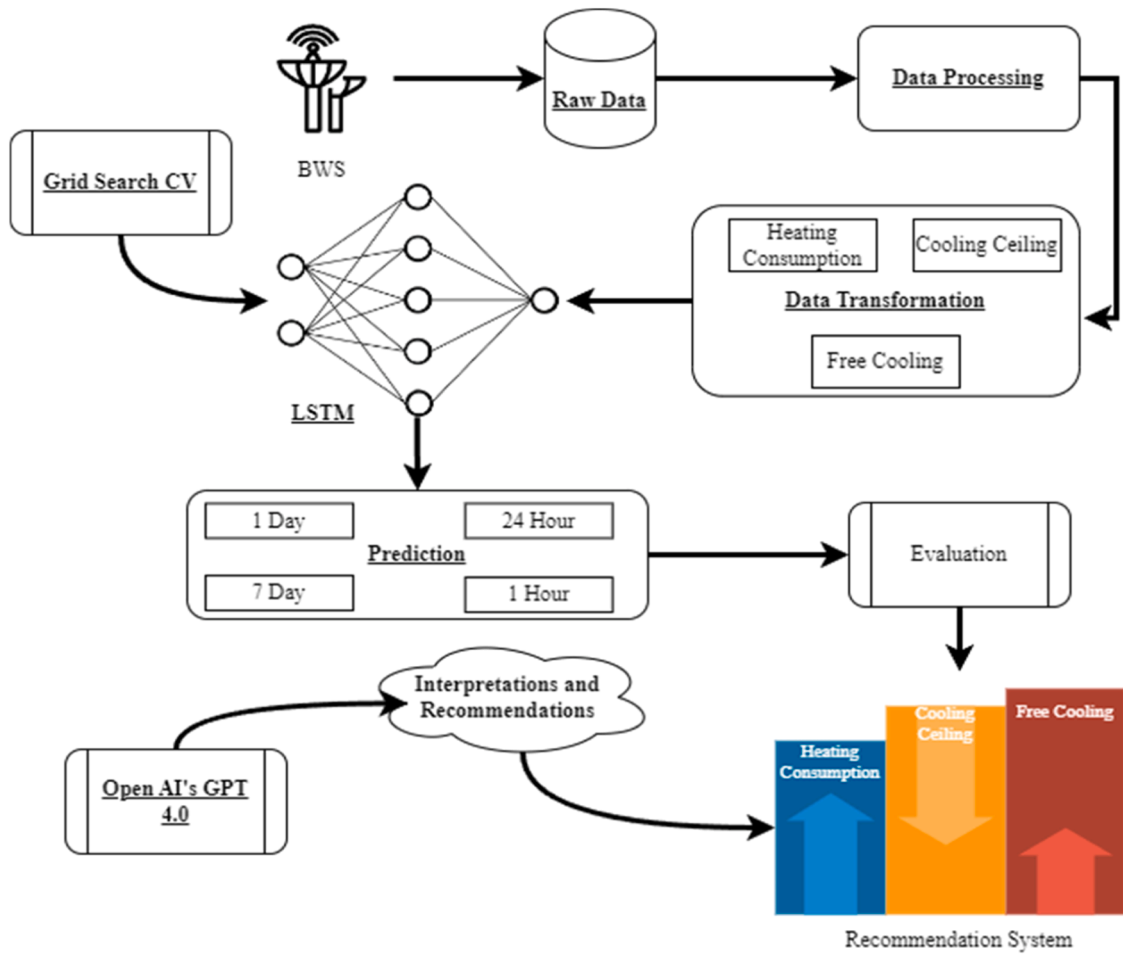


Fig. 5. Proposed methodology.

Table 2
Proposed model architecture.

Layer Type	Number of Units	Activation Function	Purpose
LSTM / GRU (1st layer)	150	tanh/relu	Captures long-term dependencies in sequences
LSTM / GRU (2nd layer)	150	tanh/relu	Refines extracted temporal features
Dense (Output)	3	ReLU	Outputs efficiency levels for CC, FC, and SH

Table 3
Optimal hyperparameters for LSTM/GRU (via GridSearchCV).

Parameter	Best Value
Units	150
Activation Function	relu
Optimizer	adam
Learning Rate	0.001
Epochs	20
Batch Size	16
Cross-Validation	3-fold

(MSE) during training. Additional metrics used for evaluation include Mean Absolute Error (MAE) and R^2 (coefficient of determination), to assess both accuracy and model fit.

Table 4
Hyperparameter ranges for XGBoost (via RandomizedSearchCV).

Parameter	Range
n_estimators	[100, 200, 500, 1000]
learning_rate	[0.01, 0.05, 0.1, 0.2]
max_depth	[3, 5, 7, 9]
subsample	[0.6, 0.8, 1.0]
colsample_bytree	[0.6, 0.8, 1.0]
gamma	[0, 0.1, 0.2]
reg_alpha	[0, 0.1, 1.0]
reg_lambda	[1.0, 5.0, 10.0]
Cross-Validation	3-fold

4.4. Integration of GPT-4 into the recommendation system

GPT-4 was integrated as a post hoc interpretability module to enhance the explainability of control strategies derived from an LSTM-based temperature forecasting model. It was **not** used during training, forecasting, or evaluation. Instead, it interpreted statistical summaries of HVAC operation for three systems:

- Space Heating (SH)
- Cooling Ceilings (CC)
- Free Cooling (FC)

4.4.1. Technical workflow

The integration of GPT-4 into the recommendation system followed a

multi-stage technical workflow designed to enhance the interpretability of LSTM-based temperature forecasts. In the prediction stage, the LSTM model generated indoor temperature forecasts, forming the primary input for subsequent analysis. This was followed by a statistical post-processing stage, where activation thresholds for each HVAC system were computed dynamically using percentile-based logic, and operational intensity bands were calculated through the Interquartile Range (IQR) method to capture typical operating bounds.

Once these metrics were derived, the data structuring stage organized the outputs by season and occupancy type (weekday or weekend), ensuring that patterns and variations in system performance could be systematically analyzed. In the final interpretation stage, this structured dataset was passed to GPT-4, which performed three core tasks: providing natural language interpretations of system operation, identifying inefficiencies such as overlapping heating and cooling thresholds, and delivering actionable recommendations to refine the control logic for improved energy efficiency and occupant comfort. The example prompt structure of GPT-4 is shown in Table 5.

Example GPT-4 Output (Winter, excerpt)

On weekdays, SH activates below 18 °C with intensity up to 92.79 %, ensuring comfort but increasing energy use. A reduction to ~17 °C may improve efficiency. CC remains inactive as appropriate for winter. FC activates below 16 °C with up to 36.44 % intensity; consider narrowing control bands to improve stability.

4.5. Graph generation and interpretation

In the proposed methodology, graphical analysis plays a key role in translating LSTM model predictions into actionable strategies for energy management. Normalized consumption data for ceiling cooling, free cooling, and heating systems is plotted against BWS air temperature across different seasons and occupancy types, with dynamic thresholds and usage intensity bands overlaid. These visualizations reveal system behavior under varying conditions, enabling targeted operational adjustments. GPT-4 is then applied to interpret these graphs, identifying optimal operating ranges, detecting anomalies such as overlapping heating and cooling thresholds, and providing context-specific recommendations. This streamlined process transforms complex model outputs into clear, practical insights that support informed decision-making and improved energy efficiency.

4.5.1. Role of GPT-4 beyond visual summaries

GPT-4 extends beyond the role of generating visual summaries by converting multi-dimensional time series outputs into structured, context-specific recommendations. While a human expert might focus on immediately visible trends in graphical outputs, GPT-4 can process and cross-reference multiple variables simultaneously, uncovering inefficiencies or operational nuances that are not readily apparent. For instance, it can correlate high ambient temperatures with system boundary positions to flag conditions where energy usage is suboptimal despite comfort thresholds being met.

4.6. Example: human expert: “Temperature is around 22 °C; heating is high”

GPT-4: “Although the comfort range is respected, the temperature is

consistently at the upper boundary, leading to higher heating demand. A lower setpoint strategy could reduce energy consumption without compromising comfort.”

5. Results

In this study, we evaluate the performance of a recommendation system designed to predict the efficiency levels of space heating, ceiling cooling, and free cooling systems. Our approach leverages LSTM models to forecast energy consumption across different temporal scales. Specifically, we categorize our predictions into two primary groups based on data aggregation: hourly and daily. For hourly predictions, we examine two scenarios: 1 h ahead (using the preceding 24 h of data) and 24 h ahead (using a 48 h sequence). In the daily predictions, we consider both 1-day and 7-day horizons, providing insights into short-term and weekly energy planning.

5.1. Hourly

5.1.1. 1-hour prediction

In the 1 h energy consumption prediction task, the performance of three models, XGBoost (XGB), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM) was evaluated across three systems: space heating, ceiling cooling, and free cooling (Table 6). XGB consistently outperformed the deep learning models in terms of accuracy, achieving an R^2 of 0.973 for space heating, 0.962 for ceiling cooling, and 0.947 for free cooling, with correspondingly low RMSE and MAE values. These results indicate that XGB was particularly effective in modeling short-term energy demand patterns, even though it required significantly less training time and memory compared to GRU and LSTM.

The LSTM and GRU models also demonstrated strong predictive capabilities, with R^2 values exceeding 0.92 across all systems. LSTM achieved slightly better accuracy in space heating than GRU, while both models performed similarly for cooling systems. Despite their longer training durations and higher memory consumption, the deep learning models captured the temporal dependencies effectively, particularly in the case of free cooling, where RMSE and MAE values were lowest. These results support the use of LSTM and GRU for scenarios that prioritize temporal learning, while XGB remains a strong baseline for real-time applications with limited computational resources.

Table 6
Evaluation of models in 1 h energy prediction.

Model	Training Time	Memory	Space Heating Consumption (R^2 / RMSE / MAE)	Cooling Ceilings (R^2 / RMSE / MAE)	Free Cooling (R^2 / RMSE / MAE)
XGB	6.25 s	41.0 MiB	0.973/0.029/0.014	0.962/0.032/0.014	0.947/0.027/0.015
GRU	292.9 s	141.3 MiB	0.962/0.186/0.107	0.949/0.052/0.025	0.926/0.019/0.010
LSTM	369.6 s	302.3 MiB	0.969/0.168/0.097	0.948/0.053/0.025	0.924/0.019/0.010

Table 5
Example prompt structure.

Prompt Section	Purpose	Example Variables
Role definition	Define GPT-4's task in interpreting post-processed HVAC performance.	"You are assisting in the interpretation of the energy performance..."
Context	State that values originate from an LSTM forecasting model.	{season}, {threshold_percentiles heating}
Data Presentation	List thresholds, IQR bands, and intensities per system and day type.	{heating_iqr_lb_weekend}, {cooling_threshold_weekday}
Task Specification	Request concise interpretation, technical observations, and actionable recommendations.	"Please analyze this information and provide..."

• Training and Validation Performance Comparison (1 h)

Fig. 6 presents the training and validation loss curves for one-hour ahead energy consumption prediction using LSTM, GRU, and XGBoost models. The LSTM and GRU networks exhibit a typical convergence pattern, with the training and validation loss decreasing steadily over epochs. Notably, GRU achieves slightly faster convergence compared to LSTM, but both models maintain comparable final validation loss levels, indicating stable learning behavior without significant overfitting. The XGBoost model, visualized through the MSE loss over boosting rounds, demonstrates rapid convergence within the first 50 rounds, after which the validation loss plateaus, suggesting that additional boosting yields minimal gains. Overall, the results indicate that all three models are capable of capturing short-term energy dynamics effectively, with deep learning models offering more consistent performance trends across epochs.

• Residual Distribution for 1 h Prediction Models

The Fig. 7 illustrates the residual distributions for three types of energy consumption, space Heating Consumption, Cooling Ceilings, and Free Cooling, across three different models. Each subplot displays the difference between actual and predicted values (residuals), with a vertical red dashed line marking the zero-error point. All distributions are tightly centered around zero, indicating that the models have low bias and generally accurate predictions. However, Space Heating Consumption exhibits slightly wider tails, suggesting higher variance or occasional large prediction errors compared to the other two systems. Cooling Ceiling and Free Cooling show narrower residual spreads, implying more consistent model performance.

5.1.2. 24-hour prediction

The evaluation of the proposed models for 24 h ahead prediction of space heating consumption shows that LSTM consistently outperforms both XGBoost (XGB) and GRU across most hours, particularly in later

hours of the day where long-term dependencies become more pronounced (Table 7). LSTM achieves the highest mean R^2 score of 0.9384 with the lowest average RMSE (0.0441) and MAE (0.0295), indicating superior predictive accuracy and error minimization over the full prediction horizon. GRU performs comparably with a mean R^2 of 0.934 and slightly higher error values, while XGB, despite its faster execution and lower resource usage (Train time: 279.89 s, memory usage: 204.8 MiB), trails with a lower mean R^2 of 0.9067 and higher average error metrics. These results highlight the effectiveness of deep learning models, especially LSTM, in capturing the temporal dynamics of energy consumption across extended forecast periods.

In the 24 h prediction of cooling ceiling energy consumption, the LSTM model demonstrates the highest overall accuracy among the three models, achieving a mean R^2 of 0.9153 along with the lowest average RMSE (0.0487) and MAE (0.0254). GRU follows closely with a mean R^2 of 0.8969, performing better than XGB, which recorded a lower mean R^2 of 0.8806 and higher error metrics. While XGB initially shows competitive performance in the early hours, its accuracy declines more sharply across the later hours. LSTM maintains consistent performance throughout the 24-hour horizon, capturing the temporal patterns of cooling ceiling operations more effectively. Despite the additional computational cost (training time: 364.6 s, memory usage: 136.0 MiB), LSTM offers a more robust solution for day-ahead cooling prediction, particularly valuable in systems requiring stability and precision over extended periods as shown in Table 8.

In the 24 h prediction of free cooling energy consumption, the LSTM model delivers the lowest average RMSE (0.0339) and MAE (0.019), indicating superior error performance compared to both GRU and XGBoost (XGB) (Table 9). However, LSTM records a lower mean R^2 of 0.7676, reflecting reduced explanatory power, particularly during the later hours of the day. GRU performs moderately with a mean R^2 of 0.8232, while XGB achieves the highest mean R^2 of 0.846 but at the cost of higher prediction errors. Notably, LSTM excels in the initial hours, reaching an R^2 of 0.9426 in Hour 1, but its performance degrades progressively as the prediction horizon extends. Despite having the smallest

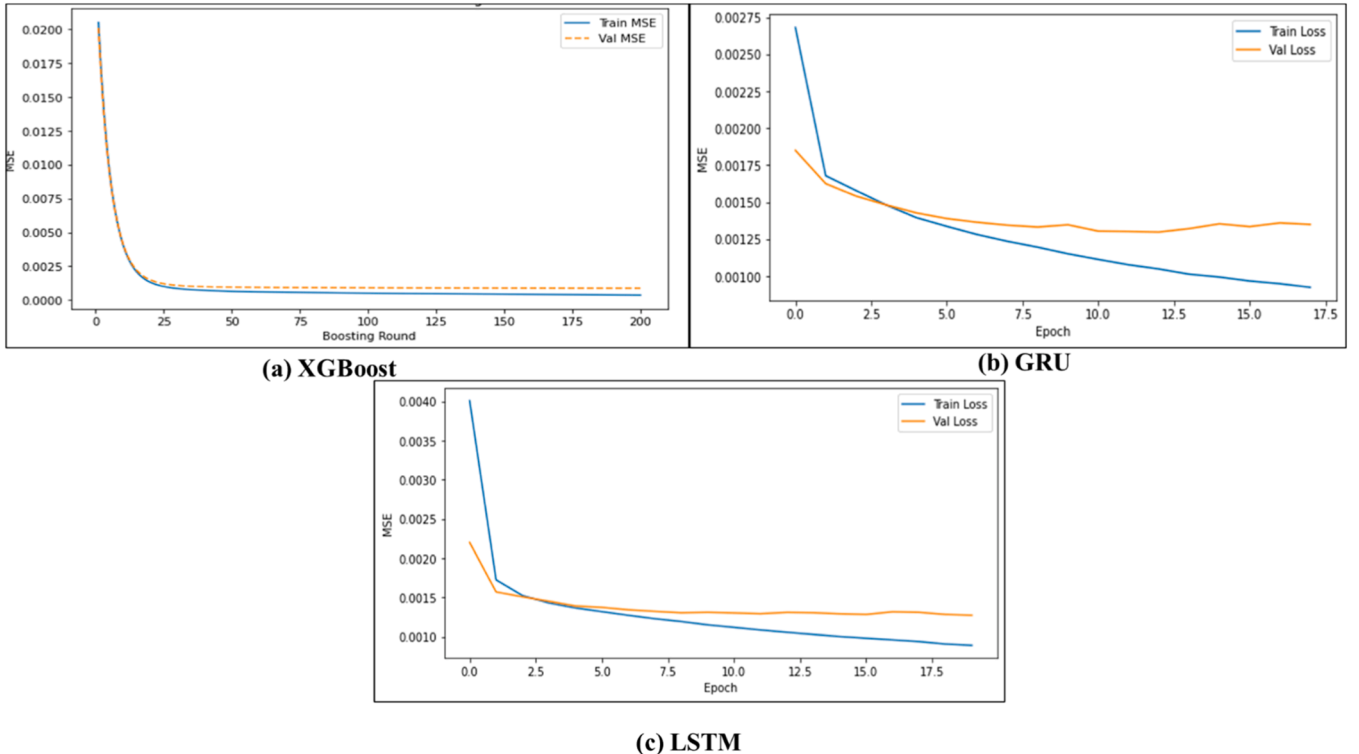


Fig. 6. Training and validation performance of models in 1 h.

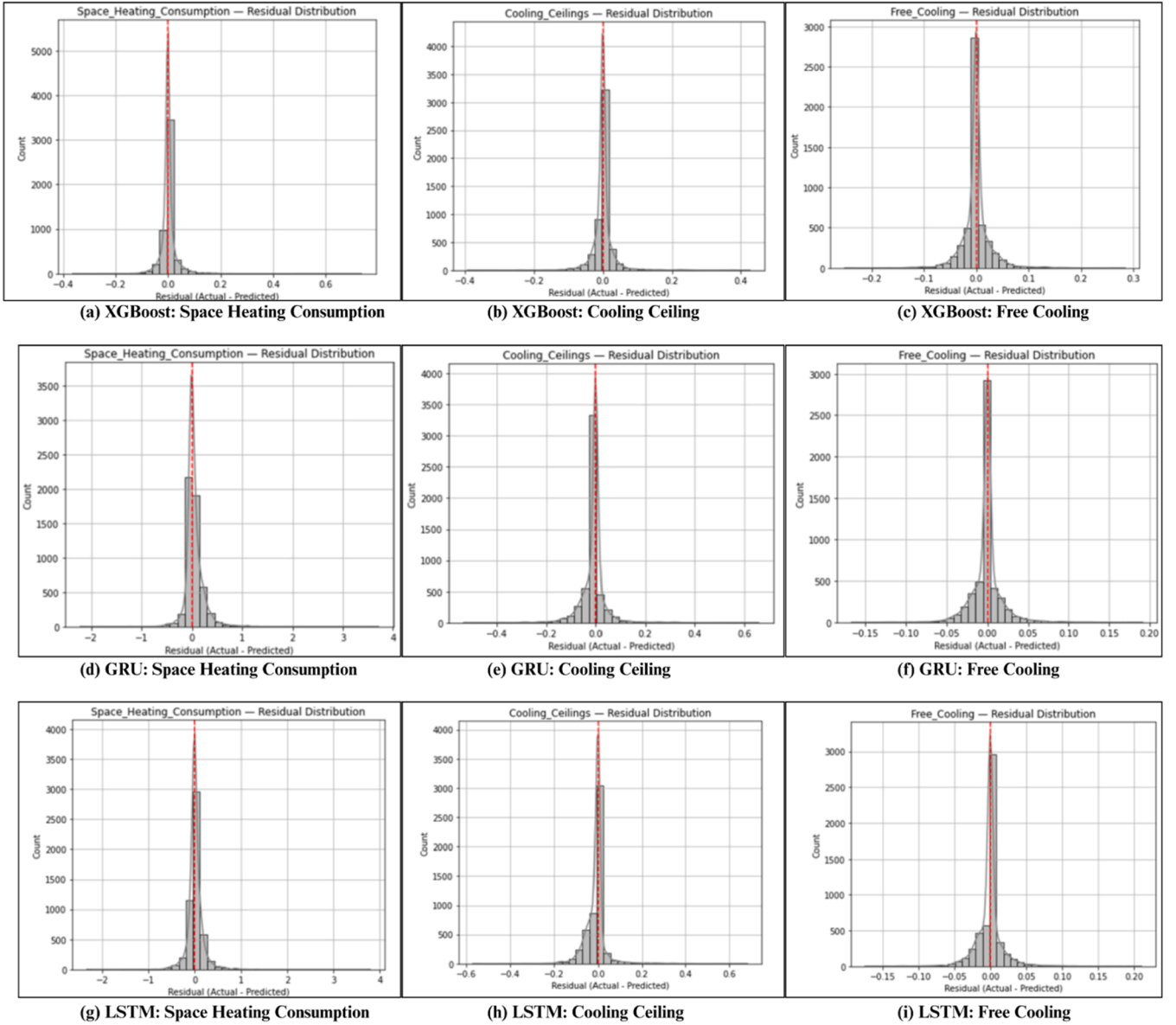


Fig. 7. Residual distributions of predicted vs. actual values (1 h).

memory footprint (71.8 MiB) and moderate compute time (Training time: 308.5 s), the results suggest that while LSTM minimizes prediction errors, its ability to explain the variance in free cooling consumption weakens over time, possibly due to the complex and less consistent nature of this subsystem's operational dynamics.

• Training and Validation Performance Comparison (24 h)

The training and validation loss curves for the 24 h ahead prediction task indicate that all three models, XGBoost, GRU, and LSTM, exhibited stable convergence behavior without signs of overfitting (Fig. 8). XGBoost, optimized over 160 boosting rounds, demonstrated a smooth decline in both training and validation mean squared error (MSE), with the two curves converging closely. GRU and LSTM also showed consistent reductions in MSE over the course of training epochs, with both models achieving similar validation performance. The marginal difference between training and validation losses across all models suggests good generalization capability.

• Residual Distribution for 24 H Prediction Models

The Fig. 9 presents residual distributions for 24 h ahead predictions of Space Heating Consumption, Cooling Ceilings, and Free Cooling across three different modeling configurations. Each subplot visualizes the spread of prediction errors (actual minus predicted), with the red dashed line representing zero residual. All distributions are sharply peaked and symmetric around zero, indicating that the models maintain strong accuracy with minimal bias over extended time horizons. The Cooling Ceilings and Free Cooling models exhibit tighter residual spreads compared to Space Heating, which shows occasional long-tailed errors, suggesting higher variability or outlier events in heating demand.

5.2. Daily

5.2.1. 1-Day

The one-day ahead prediction results highlight the superior performance of the LSTM model across all three systems: space heating, cooling ceilings, and free cooling. LSTM achieves the highest R^2 values, particularly for space heating (0.969) and cooling ceilings (0.947), indicating strong explanatory power. Its RMSE and MAE values are also the lowest for space heating (0.031 and 0.018, respectively) and cooling

Table 7

Evaluation of proposed model in 24 h energy prediction (space heating).

Hour	XGB			GRU			LSTM		
	Space Heating			Space Heating			Space Heating		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.9739	0.0783	0.1557	0.961	0.025	0.0354	0.9523	0.0263	0.0392
2	0.9523	0.1122	0.2102	0.9452	0.0277	0.0419	0.9484	0.0264	0.0407
3	0.949	0.1268	0.2169	0.9461	0.028	0.0415	0.9488	0.0271	0.0405
4	0.94	0.1406	0.2357	0.945	0.0289	0.042	0.9445	0.0283	0.0422
5	0.9313	0.1511	0.2527	0.9452	0.0287	0.042	0.9461	0.0275	0.0417
6	0.9217	0.1617	0.2698	0.9439	0.0295	0.0425	0.9431	0.0284	0.0428
7	0.917	0.1659	0.276	0.9446	0.0286	0.042	0.9436	0.0286	0.0424
8	0.9113	0.1741	0.2839	0.9437	0.0286	0.0421	0.9438	0.0283	0.0421
9	0.9078	0.1775	0.2888	0.9413	0.0292	0.0429	0.9453	0.0278	0.0414
10	0.9025	0.1813	0.2961	0.9348	0.0306	0.0451	0.9383	0.029	0.0439
11	0.8982	0.1867	0.3029	0.9364	0.0306	0.0446	0.9362	0.0303	0.0447
12	0.8964	0.1913	0.3078	0.9352	0.031	0.0453	0.9409	0.0293	0.0433
13	0.8912	0.1946	0.3153	0.9361	0.0303	0.045	0.9427	0.029	0.0426
14	0.8909	0.1971	0.3162	0.9346	0.0308	0.0456	0.9399	0.0294	0.0437
15	0.8892	0.1976	0.3191	0.9359	0.0308	0.0452	0.9411	0.029	0.0433
16	0.8892	0.2013	0.3194	0.9329	0.0316	0.0463	0.9398	0.0293	0.0438
17	0.8919	0.1991	0.3154	0.9318	0.0316	0.0466	0.9395	0.0293	0.0439
18	0.8899	0.2019	0.3182	0.9304	0.0323	0.0471	0.941	0.0293	0.0434
19	0.888	0.2044	0.3192	0.9237	0.0336	0.049	0.9379	0.0299	0.0443
20	0.8827	0.2067	0.3263	0.9232	0.0339	0.0492	0.9326	0.031	0.046
21	0.8845	0.2054	0.3251	0.9189	0.0345	0.0507	0.9292	0.0317	0.0474
22	0.8874	0.205	0.3221	0.9155	0.0351	0.0519	0.9241	0.0331	0.0492
23	0.8863	0.2056	0.3225	0.9089	0.036	0.0537	0.9159	0.0343	0.0516
24	0.8876	0.2046	0.3211	0.9011	0.037	0.0561	0.9071	0.0356	0.0544
Mean	0.9067	0.178	0.289	0.934	0.031	0.0456	0.9384	0.0295	0.0441

Table 8

Evaluation of proposed model in 24 h energy prediction (cooling ceiling).

Hour	XGB			GRU			LSTM		
	Cooling Ceiling			Cooling Ceiling			Cooling Ceiling		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.9588	0.0206	0.047	0.9407	0.0201	0.0405	0.9267	0.0224	0.0451
2	0.9342	0.0284	0.0602	0.9274	0.0225	0.0455	0.9245	0.0237	0.0464
3	0.9235	0.0316	0.0654	0.9161	0.0239	0.0493	0.9246	0.0236	0.0467
4	0.9152	0.034	0.0689	0.9134	0.0245	0.0501	0.924	0.0237	0.0469
5	0.9066	0.0374	0.0721	0.9162	0.0251	0.0491	0.9243	0.0246	0.0467
6	0.8988	0.0392	0.0745	0.9108	0.0253	0.0503	0.9144	0.0258	0.0493
7	0.8914	0.0409	0.0769	0.909	0.0259	0.0506	0.9187	0.0248	0.0478
8	0.8873	0.0425	0.0786	0.9091	0.0264	0.0508	0.9222	0.0245	0.047
9	0.8817	0.0442	0.0805	0.903	0.0269	0.0525	0.918	0.025	0.0482
10	0.8793	0.0453	0.0812	0.9043	0.0277	0.052	0.921	0.0253	0.0473
11	0.8707	0.0461	0.084	0.9001	0.0274	0.0531	0.9194	0.0253	0.0477
12	0.8661	0.047	0.0852	0.895	0.0276	0.0543	0.9189	0.025	0.0477
13	0.8657	0.0475	0.0852	0.8907	0.0283	0.0553	0.9144	0.0257	0.0489
14	0.8609	0.0483	0.0867	0.8861	0.0294	0.0564	0.9167	0.0251	0.0482
15	0.8613	0.0485	0.087	0.8875	0.0292	0.0564	0.9131	0.0262	0.0495
16	0.853	0.0494	0.0896	0.8843	0.0299	0.0572	0.9146	0.025	0.0491
17	0.8539	0.0488	0.0884	0.8916	0.0288	0.0547	0.9172	0.025	0.0478
18	0.8616	0.0476	0.0861	0.8825	0.0297	0.0571	0.9146	0.0257	0.0487
19	0.8629	0.0479	0.0852	0.8848	0.0294	0.0562	0.9135	0.0258	0.0487
20	0.866	0.0479	0.0843	0.8812	0.0295	0.0571	0.9096	0.0261	0.0498
21	0.8611	0.0487	0.0858	0.8808	0.0301	0.0572	0.9085	0.0271	0.0501
22	0.8591	0.0493	0.087	0.8763	0.0315	0.0586	0.9028	0.0277	0.052
23	0.8595	0.0485	0.0861	0.8723	0.0318	0.0591	0.8988	0.028	0.0526
24	0.8561	0.0498	0.0883	0.8617	0.0336	0.0623	0.8859	0.0297	0.0565
Mean	0.8806	0.0433	0.0798	0.8969	0.0277	0.0536	0.9153	0.0254	0.0487

ceilings (0.039 and 0.018), demonstrating its ability to generate highly accurate predictions. For free cooling, LSTM maintains competitive performance with an R² of 0.905 and low error values, though slightly behind GRU in variance explanation. This demonstrates LSTM's strength in learning complex temporal dependencies in building energy patterns over a full 24 h period.

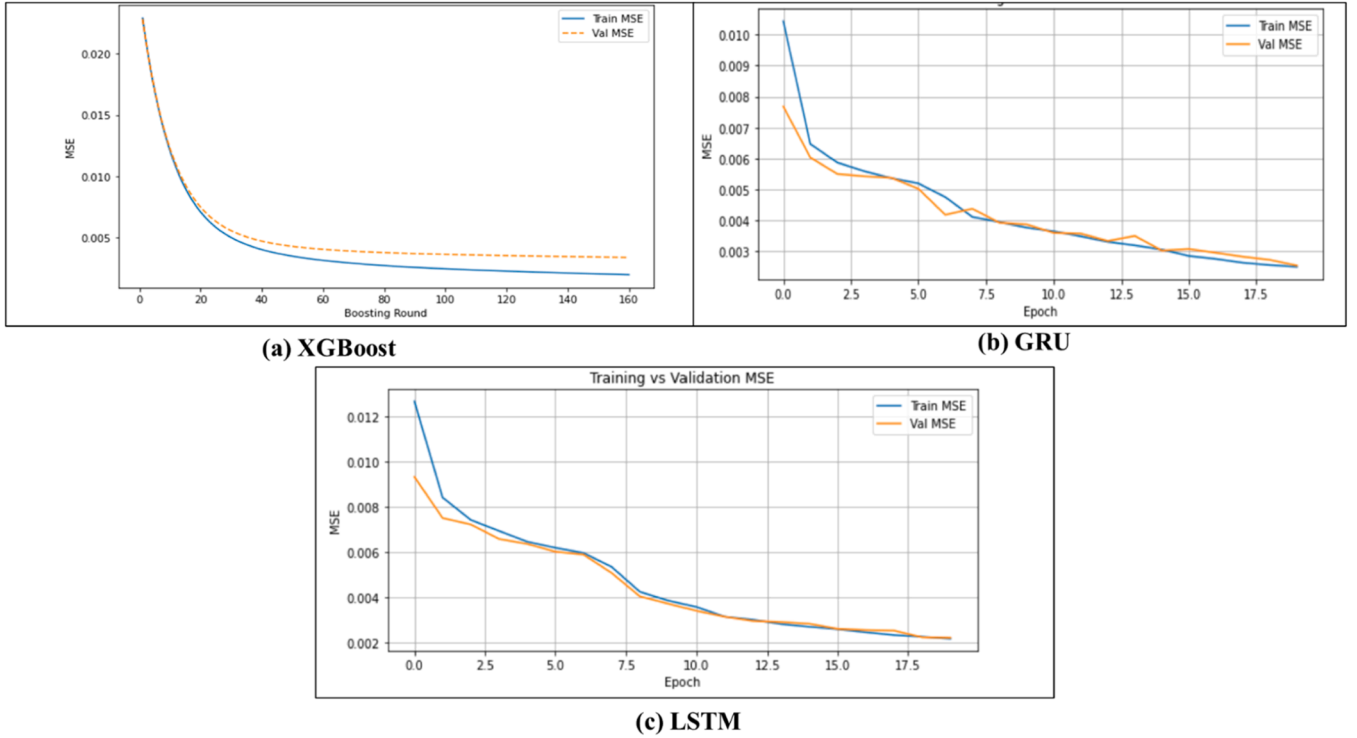
The GRU model also performs well, closely following LSTM, particularly for space heating (R² = 0.967) and cooling ceilings (R² = 0.945). It surpasses LSTM slightly in free cooling R² (0.920 vs. 0.905), although

with marginally higher error metrics. XGBoost, while significantly faster to train and requiring minimal memory, shows notably lower performance across all metrics, with R² values under 0.88 for space heating and even lower for cooling systems. These results suggest that although XGBoost is computationally efficient, it may be insufficient for capturing the temporal complexity inherent in building energy consumption, whereas LSTM, despite higher training demands, offers the most accurate and consistent predictions across systems (Table 10).

Table 9

Evaluation of proposed model in 24 h energy prediction (free cooling).

Hour	XGB			GRU			LSTM		
	Free Cooling			Free Cooling			Free Cooling		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.8755	0.0227	0.0421	0.9046	0.0201	0.0368	0.9426	0.0089	0.0169
2	0.8676	0.0217	0.0434	0.8789	0.0204	0.0415	0.8788	0.0117	0.0245
3	0.8673	0.0223	0.0436	0.866	0.0222	0.0439	0.8502	0.0138	0.0274
4	0.8701	0.022	0.0434	0.8656	0.0228	0.0442	0.8422	0.0147	0.0282
5	0.8673	0.0221	0.044	0.8627	0.0232	0.0447	0.8218	0.016	0.0301
6	0.864	0.0231	0.0446	0.8537	0.0243	0.0462	0.806	0.017	0.0314
7	0.8577	0.0238	0.0455	0.8365	0.0257	0.0488	0.7864	0.0181	0.0329
8	0.8655	0.0233	0.0442	0.8309	0.0269	0.0495	0.7823	0.0185	0.0332
9	0.8565	0.0242	0.0456	0.8351	0.0262	0.0489	0.7716	0.0192	0.034
10	0.8591	0.0237	0.0453	0.8347	0.0262	0.0491	0.7683	0.0196	0.0343
11	0.8521	0.0237	0.0463	0.831	0.0266	0.0495	0.7669	0.0198	0.0343
12	0.8502	0.0246	0.0473	0.825	0.0273	0.0511	0.7607	0.0202	0.0352
13	0.8511	0.0251	0.047	0.8183	0.0278	0.052	0.7447	0.0205	0.0363
14	0.8478	0.0251	0.0478	0.8136	0.0285	0.0529	0.7256	0.0215	0.0379
15	0.8537	0.025	0.0461	0.823	0.0284	0.0508	0.7332	0.0213	0.0368
16	0.8507	0.0246	0.0467	0.8129	0.0283	0.0523	0.7253	0.0214	0.0374
17	0.8427	0.0252	0.0481	0.8034	0.0288	0.0538	0.7114	0.0218	0.0385
18	0.8384	0.0254	0.0486	0.7946	0.0289	0.0547	0.7065	0.0218	0.0386
19	0.8412	0.0256	0.0476	0.8011	0.0289	0.0533	0.7215	0.0216	0.0372
20	0.8292	0.0257	0.0497	0.7918	0.0292	0.0548	0.723	0.0212	0.0373
21	0.8109	0.0276	0.0522	0.7846	0.0297	0.0557	0.7197	0.0214	0.0375
22	0.8105	0.0281	0.0521	0.7806	0.0311	0.056	0.7184	0.0218	0.0375
23	0.8054	0.0288	0.0524	0.7676	0.0318	0.0572	0.7161	0.0218	0.0373
24	0.77	0.0313	0.0575	0.7395	0.0337	0.0612	0.6998	0.0223	0.0387
Mean	0.846	0.0248	0.0471	0.8232	0.0269	0.0504	0.7676	0.019	0.0339

**Fig. 8.** Training and validation performance of models in 24 h.

• Training and Validation Performance Comparison (1 day)

The 1-day ahead prediction results indicate that all three models, XGBoost, GRU, and LSTM, achieved effective convergence with minimal overfitting (Fig. 10). XGBoost exhibited a steady and consistent decline in training and validation mean squared error (MSE) across 550 boosting rounds, with the curves closely aligned, indicating strong

generalization. GRU showed a smooth reduction in both training and validation losses, with a small and stable gap between the two, reflecting controlled variance during learning. LSTM similarly demonstrated rapid loss minimization within the first few epochs, followed by stable performance across subsequent iterations. The small divergence between training and validation losses across all models confirms that the models are not overfitting and are well-suited for short-term energy

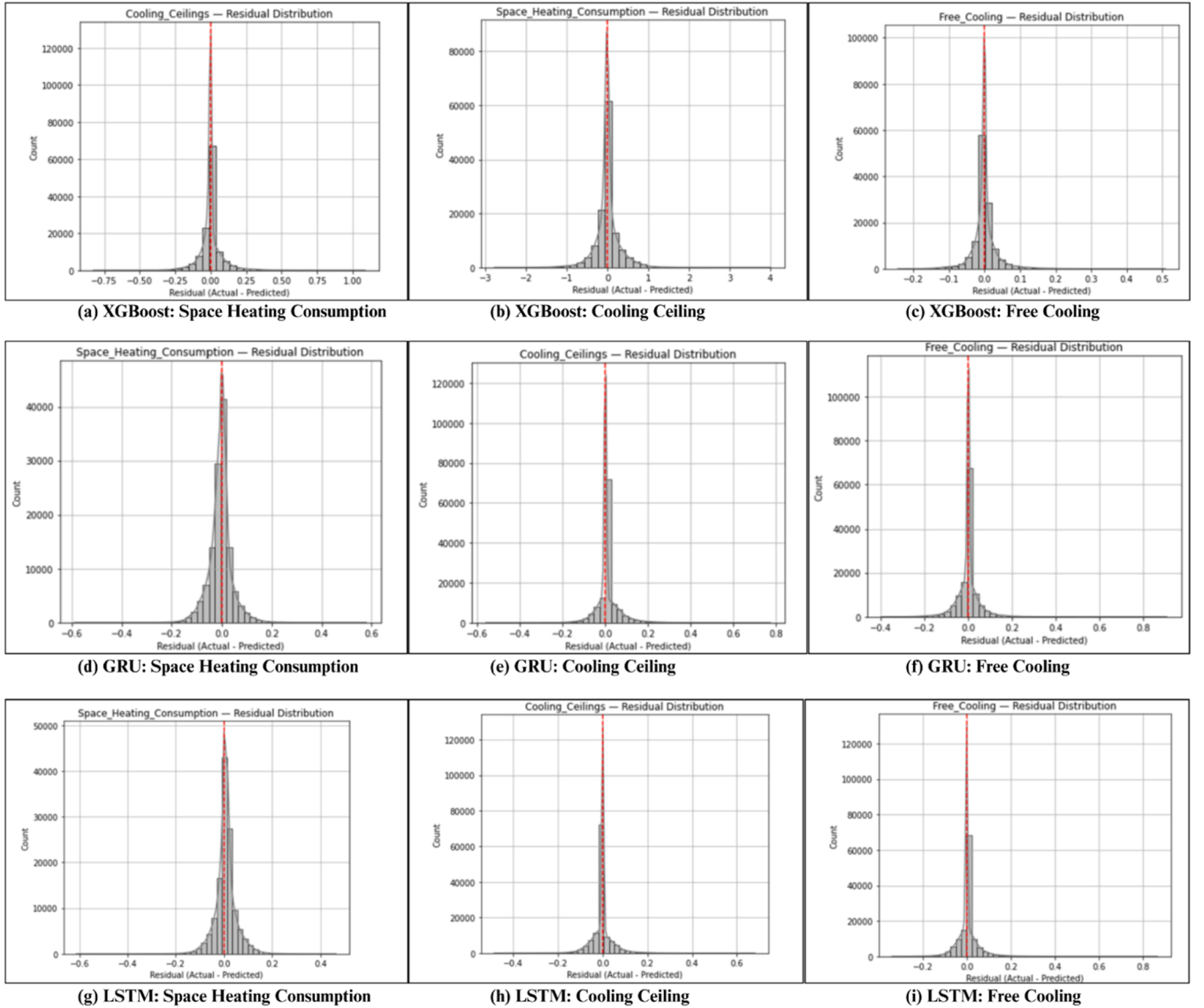


Fig. 9. Residual distributions of predicted vs. actual values (24 H).

Table 10
Evaluation of proposed model in 1-day energy prediction.

Model	Training Time	Memory	Space Heating Consumption (R ² / RMSE / MAE)	Cooling Ceilings (R ² / RMSE / MAE)	Free Cooling (R ² / RMSE / MAE)
XGBoost	19.2 s	3.6 MiB	0.879/0.061/0.039	0.846/0.066/0.036	0.795/0.056/0.031
GRU	271.1 s	200.0 MiB	0.967/0.171/0.106	0.945/0.055/0.022	0.920/0.021/0.009
LSTM	294.1 s	74.1 MiB	0.969/0.031/0.018	0.947/0.039/0.018	0.905/0.038/0.017

consumption forecasting.

• Residual Distribution for 1 Day Prediction Models

The Fig. 11 displays the residual distributions for 1-day ahead

predictions of Space Heating Consumption, Cooling Ceilings, and Free Cooling, across three different modeling settings. All histograms represent the difference between actual and predicted values, with a red dashed line denoting zero residual. The distributions are characteristically sharp, centered around zero, and largely symmetric, indicating a minimal prediction bias. However, Free Cooling displays slightly longer right tails in some plots, suggesting occasional underestimation. Meanwhile, Cooling Ceilings and Space Heating maintain narrower error bands, though some plots indicate minor deviations in skewness.

5.2.2. 7-Day

In the 7-day space heating prediction task, the LSTM model demonstrates strong and consistent performance, achieving a mean R² of 0.9303 with lower RMSE (0.0462) and MAE (0.0287) compared to GRU and XGBoost (XGB) (Table 11). While XGB records the highest R² values on Days 1 and 2 (0.9739 and 0.9523), its accuracy steadily declines over the week, resulting in higher average errors (MAE: 0.1431, RMSE: 0.2307). GRU performs reasonably well with a mean R² of 0.9108, though its errors are higher than those of LSTM. Notably, LSTM maintains stable prediction quality throughout all seven days, indicating better generalization and the ability to capture long-term temporal dependencies. Although LSTM and GRU require more training time and

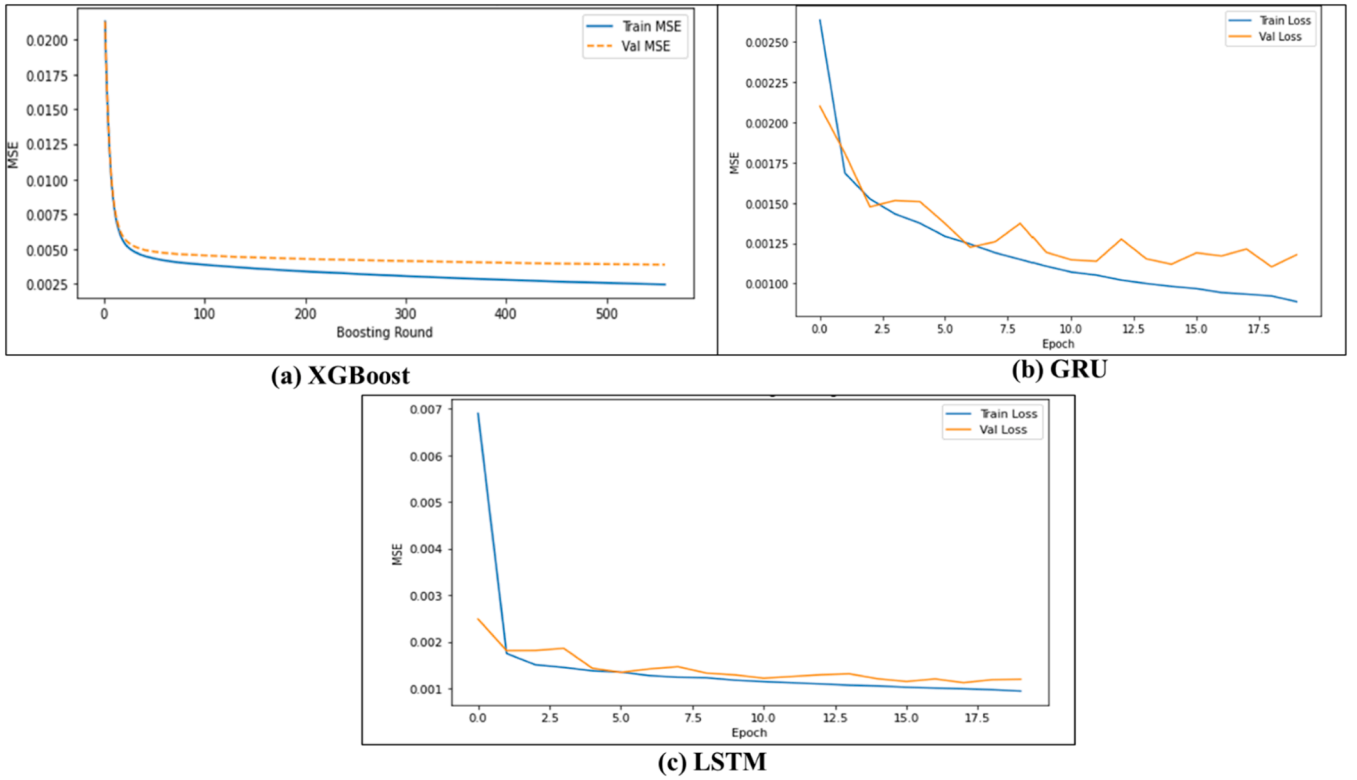


Fig. 10. Training and validation performance of models in 1 day.

memory compared to XGB, their improved predictive reliability makes them more suitable for multi-day energy forecasting in dynamic building environments.

For the 7-day prediction of cooling ceiling energy consumption, the LSTM model demonstrates the best overall performance with the lowest mean RMSE (0.0515) and competitive MAE (0.0258), along with the highest average R^2 (0.9036), indicating strong predictive accuracy and consistency (Table 12). GRU closely follows, achieving a similar RMSE (0.0519) and slightly lower MAE (0.0250), with a mean R^2 of 0.9021. XGBoost (XGB), while performing well on Day 1 with an R^2 of 0.9522, exhibits a steady decline in performance across the week, culminating in a lower mean R^2 of 0.8936 and notably higher error values (MAE: 0.0413, RMSE: 0.075). This suggests that deep learning models, particularly LSTM, are better suited for capturing the temporal dependencies inherent in cooling ceiling operations over an extended forecasting horizon. While training times and memory usage are higher for LSTM (108.3 s, 58.1 MiB) and GRU (92.4 s, 71.5 MiB) compared to XGB (9.94 s, 93.1 MiB), the improved forecasting accuracy justifies their application in predictive control scenarios.

In the 7-day prediction of free cooling energy consumption, the LSTM model consistently outperforms GRU and XGBoost in terms of overall accuracy and error metrics. It achieves the highest average R^2 value of 0.8364, indicating stronger explanatory power across the forecast horizon. Additionally, LSTM records the lowest mean RMSE (0.049) and a competitive MAE (0.0245), reflecting its ability to make precise predictions with reduced variance. While GRU trails slightly behind with an R^2 of 0.8102 and a marginally higher RMSE (0.0525), XGBoost shows a notable decline in predictive accuracy beyond Day 1, resulting in the lowest average R^2 (0.7901) and the highest error values (MAE: 0.0176, RMSE: 0.0324). The results confirm that recurrent neural network architectures, particularly LSTM, are better suited for modeling the temporal dynamics and non-linear dependencies associated with free cooling demand, especially over extended periods. Table 13

• Training and Validation Performance Comparison (7 days)

The training and validation results for the 7-day prediction task indicate that all three models, XGBoost, GRU, and LSTM, successfully reduced mean squared error (MSE) throughout their learning processes, reflecting effective model training and generalization (Fig. 12). The GRU and LSTM models exhibited consistent loss trajectories between training and validation, suggesting stable learning behavior with minimal signs of overfitting. Similarly, XGBoost maintained low and parallel MSE values for both phases, confirming its robustness. Among the three, LSTM achieved the lowest validation MSE, highlighting its strength in capturing complex temporal dependencies in building energy consumption data.

• Residual Distribution for 1 Day Prediction Models

The residual distribution plots for the 7-day prediction models, XGBoost, GRU, and LSTM, across space heating, cooling ceiling, and free cooling systems reveal how well each model captures the variations between actual and predicted energy consumption (Fig. 12). All models exhibit residuals densely concentrated near zero, indicating strong predictive accuracy. Notably, the LSTM model shows the tightest clustering of residuals around the zero line, reflecting its superior ability to handle nonlinear and temporal dependencies in the data with minimal bias. The GRU model performs similarly but with slightly wider spread in residuals, and XGBoost, while effective, displays slightly heavier tails, suggesting occasional larger prediction errors.

5.3. Recommendation system

In this section, a detailed analysis of the recommendation system is presented. The proposed recommendation system will improve energy efficiency and reduce energy costs to manage space heating and cooling in various seasonal and occupancy conditions. The system employs six strategies tailored to specific conditions, including efficiently scaling heating intensity on weekends, tracking outliers through interquartile range (IQR) analysis, and setting outside trigger temperatures based on

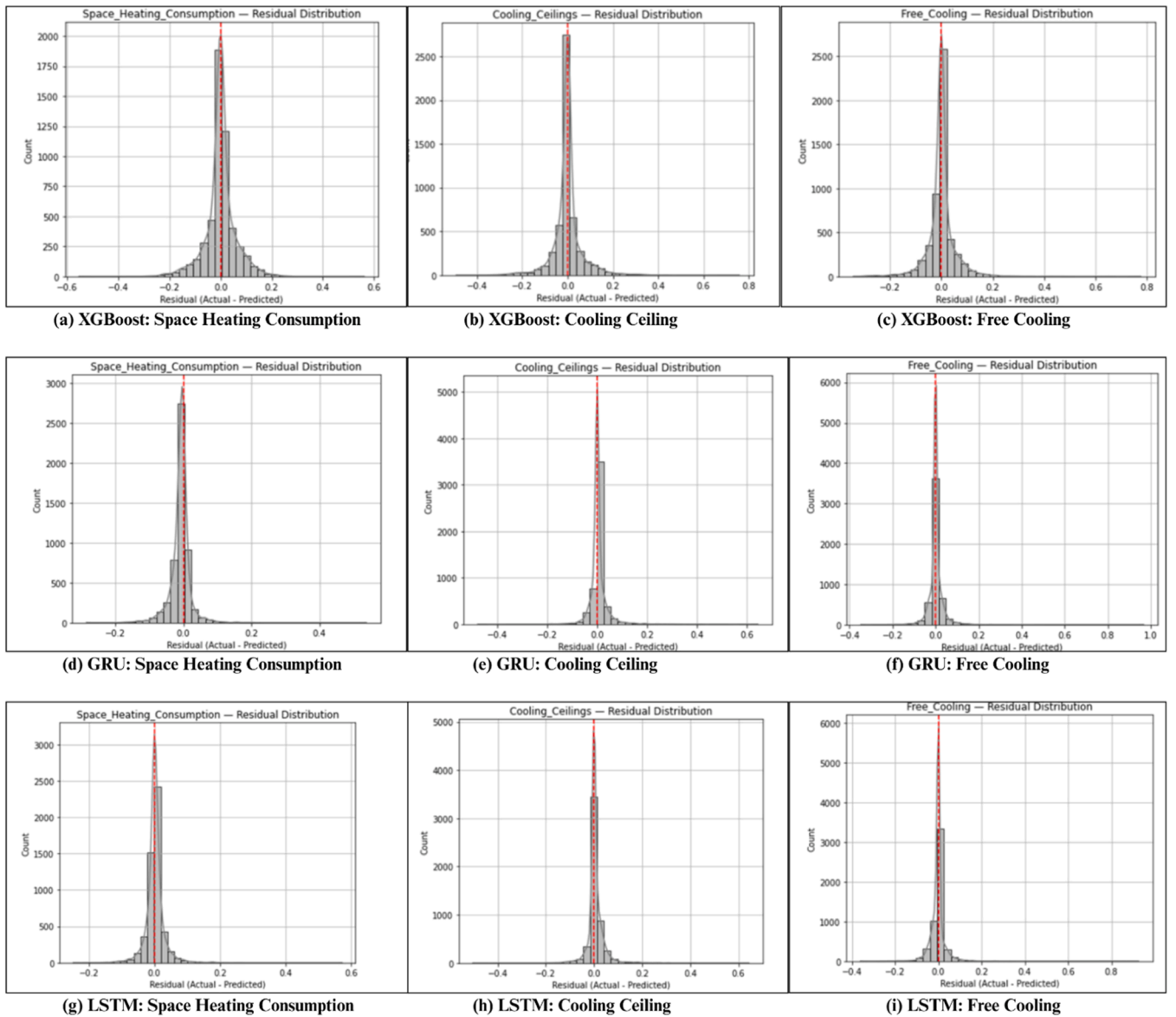


Fig. 11. Residual distributions of predicted vs. actual values (1 Day).

Table 11

Evaluation of proposed model in 7 day energy prediction (space heating).

Day	XGB			GRU			LSTM		
	Space Heating			Space Heating			Space Heating		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.9739	0.0819	0.1552	0.97	0.0191	0.031	0.9685	0.0197	0.0318
2	0.9523	0.1246	0.2087	0.94	0.0267	0.0436	0.9512	0.0241	0.0393
3	0.9379	0.1451	0.2378	0.9166	0.031	0.0513	0.938	0.0272	0.0442
4	0.9342	0.1547	0.2449	0.9	0.0338	0.0562	0.93	0.0289	0.0471
5	0.9308	0.1605	0.2509	0.8934	0.0356	0.058	0.9195	0.0313	0.0504
6	0.9289	0.1646	0.253	0.8859	0.0369	0.0597	0.9084	0.0336	0.0535
7	0.9226	0.1704	0.2644	0.8701	0.0396	0.0638	0.8966	0.0359	0.0569
Mean	0.9401	0.1431	0.2307	0.9108	0.0318	0.0519	0.9303	0.0287	0.0462

high percentiles (95th for weekends and 100th for weekdays). The IQR, representing the range between the 25th and 75th percentiles, helps identify unusual energy consumption patterns, allowing for adjustments to avoid inefficiencies. Percentiles divide a dataset into 100 equal parts, and by using high percentiles for trigger temperatures, the system can respond effectively to extreme conditions, conserving energy while

maintaining comfort. For cooling, the system works contrarily, with trigger temperatures based on the 2.5th and 0th percentiles. These strategies ensure that the system operates optimally, adjusting heating and cooling operations precisely to meet demand while conserving energy

The interpretation module, utilizing OpenAI's Generative Pretrained

Table 12

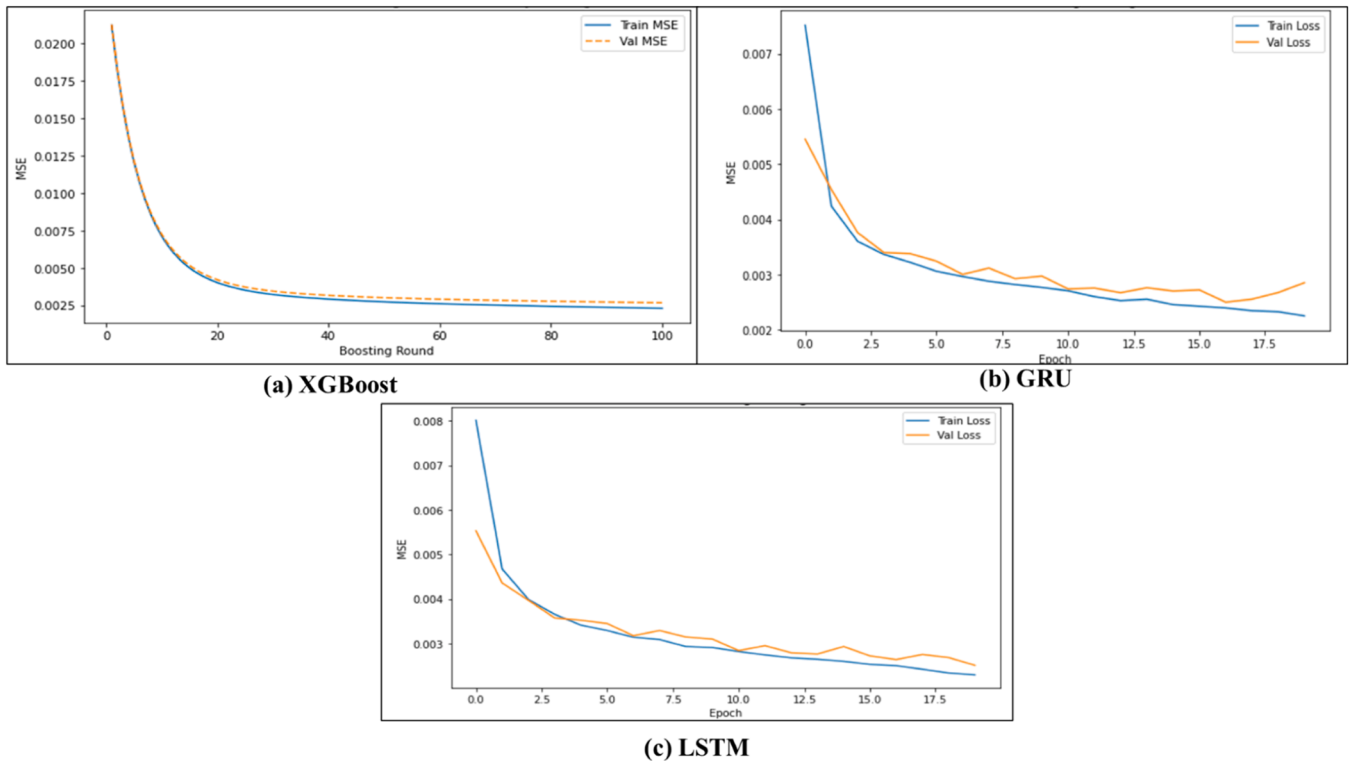
Evaluation of proposed model in 7 day energy prediction (cooling ceiling).

Day	XGB			GRU			LSTM		
	Cooling Ceiling			Cooling Ceiling			Cooling Ceiling		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.9522	0.0241	0.0512	0.9503	0.0168	0.0376	0.9468	0.0181	0.0389
2	0.9149	0.0354	0.0685	0.9208	0.0216	0.0476	0.9197	0.0226	0.0479
3	0.8978	0.0405	0.0742	0.9057	0.0238	0.0513	0.9121	0.0246	0.0495
4	0.8934	0.0424	0.0755	0.9043	0.0247	0.0515	0.9082	0.0253	0.0504
5	0.8802	0.0458	0.0803	0.8919	0.0273	0.0549	0.895	0.0273	0.0541
6	0.8602	0.0495	0.0874	0.8737	0.0295	0.0598	0.8781	0.0303	0.0587
7	0.8568	0.0516	0.0878	0.8684	0.0312	0.0606	0.8655	0.0327	0.0612
Mean	0.8936	0.0413	0.075	0.9021	0.025	0.0519	0.9036	0.0258	0.0515

Table 13

Evaluation of proposed model in 7 day energy prediction (free cooling).

Day	XGB			GRU			LSTM		
	Free Cooling			Free Cooling			Free Cooling		
	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE	(R ²)	MAE	RMSE
1	0.9295	0.0105	0.0191	0.9148	0.0186	0.0356	0.9087	0.0198	0.0369
2	0.853	0.0137	0.0276	0.8611	0.0213	0.0456	0.8639	0.0221	0.0451
3	0.8162	0.0162	0.0307	0.8349	0.0232	0.0494	0.8579	0.0224	0.0458
4	0.7678	0.0187	0.035	0.8008	0.0261	0.0549	0.8415	0.0239	0.049
5	0.7515	0.02	0.036	0.777	0.028	0.0579	0.8241	0.0256	0.0514
6	0.7202	0.0213	0.038	0.7519	0.0297	0.0607	0.7914	0.0281	0.0557
7	0.6923	0.0229	0.0401	0.7307	0.0312	0.0635	0.7677	0.0295	0.059
Mean	0.7901	0.0176	0.0324	0.8102	0.0255	0.0525	0.8364	0.0245	0.049

**Fig. 12.** Training and validation performance of models in 7 days.

4.0 model, further enhances the recommendation system by providing an in-depth analysis of graphical representations of energy consumption data. This module interprets visualized data to elucidate complex patterns and interactions, offering actionable insights that inform the adjustment strategies. For example, the GPT-4 model aids in understanding the implications of outlier detection through IQR and the

significance of high percentiles in setting trigger temperatures. This interpretation ensures that the system's recommendations are grounded in a nuanced understanding of the data, leading to more precise adjustments in heating and cooling operations.

5.3.1. Hourly space heating consumption

The chosen percentiles for space heating consumption are designed to fine-tune the system's response to temperature variations across different seasons and days of the week as shown in Table 14. During weekends, a 97.5th percentile is used, indicating that the system will activate heating when temperatures fall below the value higher than 97.5 % of the recorded data points for that season. This approach ensures heating is employed judiciously, conserving energy during typically lower occupancy periods. Conversely, on weekdays, the 100th percentile is utilized, meaning heating is triggered only at the most extreme temperature values, reflecting the higher occupancy and activity levels.

The winter weekend plot (Fig. 13) shows that the system activates space heating when the Building Weather Station (BWS) air temperature drops below 14 °C, with normalized consumption ranging from 0–50 %, ensuring indoor comfort during colder weather. The green dashed line at 55 % indicates efficient moderation of heating intensity on weekends, reflecting lower occupancy demands. On weekdays, the heating threshold increases to 18 °C due to higher occupancy and activity levels, resulting in intensified heating consumption from 0 % to nearly 100 % as temperatures fall. The upper bound for weekday heating is set at 93 %, demonstrating the system's ability to meet higher heating requirements efficiently.

The spring plots (Fig. 14) illustrate the normalized space heating consumption versus BWS air temperature for spring, segmented by weekend and weekday. Both scenarios show a clear negative correlation between space heating consumption and air temperature. During weekends, as depicted in the left plot, this represents the system's intensity, where heating consumption is higher at lower temperatures and decreases steadily, approaching zero near the 20 °C threshold (blue dashed line). The data demonstrates an effective control mechanism, as consumption rarely exceeds the 30.00 % upper bound. Similarly, on weekdays, shown in the right plot, the intensity of the system declines as temperatures rise, ceasing near the 20 °C threshold. We employ a rigid approach in the weekday scenario by considering the 100th percentile (excluding negative values). As a result, heating is still activated at temperatures as high as 28 °C. However, the intensity remains insignificant until the temperature drops below 15 °C, where it exceeds 20 %. The weekday consumption data adheres well to the 61.00 % upper bound, indicating consistent heating management. The weekday consumption data adheres well to the 61.00 % upper bound, indicating consistent heating management.

5.3.2. Hourly cooling ceiling consumption

The chosen percentiles for our analysis, specifically 2.5 % for weekends and 0 % for weekdays across all seasons (Winter, Spring, Summer, and Autumn), are critical in setting the thresholds for the hourly cooling ceiling consumption as shown in Table 15. These percentiles were adopted to ensure that the cooling system operates optimally under varying occupancy patterns, with a stricter control on weekdays when building activity is typically higher, and slightly more lenient on weekends.

The analysis of cooling ceiling consumption during summer demonstrates the effectiveness of the recommended system in managing cooling requirements across varying temperature conditions (Fig. 15). On weekends, the system activates cooling only when temperatures exceed 14 °C, aligning with energy-saving principles to minimize usage

at lower temperatures and optimize energy consumption. While most consumption levels remain within acceptable limits, some instances exceed the upper bound of 56 %, indicating periods of heightened demand that require further optimization. However, turning on the air conditioning at an outside temperature of 10 °C may not be necessary, as this can lead to unnecessary energy consumption. This phenomenon might be linked to the system's need to rebalance due to overly intense heating, which subsequently requires cooling, or it may be necessary to maintain specific technical areas at lower temperatures. On weekdays, the system effectively maintains indoor comfort when outdoor temperatures exceed 10 °C, with increased consumption during business hours. We recommend maintaining comfort levels during weekdays to accommodate occupants, while adapting the system during weekends when the building is largely unoccupied, thus offering an opportunity to save energy without compromising comfort.

In case of spring, the weekend plot (Fig. 16) shows that cooling consumption remains minimal until the temperature exceeds the 9 °C threshold, beyond which there is a significant increase in consumption, particularly between 15 °C and 25 °C. The system effectively maintains consumption within the 25 % upper bound, demonstrating efficient control. Conversely, the weekday plot indicates that cooling consumption starts to rise as temperatures surpass the 6 °C threshold, peaking significantly between 10 °C and 30 °C. Despite a broader temperature range, the weekday data adheres well to the 41 % upper bound, indicating consistent management. These observations suggest that the cooling system is adept at minimizing energy use during lower temperatures and scaling up efficiently as temperatures rise, maintaining consumption within the defined upper bounds.

On the other hand for Autumn, on weekends, the graph (Fig. 17) shows a distinct pattern where normalized cooling ceilings consumption remains relatively low and stable when the BWS air temperature is below approximately 13 °C. As temperatures rise above this threshold, there is a noticeable increase in cooling consumption, indicated by a higher dispersion of data points. The green dashed line marks the upper bound at 25 % consumption, while the purple dashed line indicates the temperature threshold at 12 °C. Similarly, weekdays, displays a similar trend but with more pronounced increases in cooling consumption as temperatures rise. Notably, there is a broader spread of data points compared to weekends, suggesting higher variability in cooling demand during weekdays. The upper bound, marked at 36 % consumption, is higher compared to the weekend, reflecting potentially higher usage or demand during working days. The temperature threshold remains consistent at 8 °C.

5.3.3. Hourly free cooling consumption

The chosen percentiles of 100 % for both weekends and weekdays across all seasons (Winter, Spring, Summer, and Autumn) indicate that there are no specific thresholds or upper limits set for the hourly cooling ceiling consumption as shown in Table 16. This approach implies that the cooling system is expected to operate continuously and at its maximum capacity regardless of the day of the week or seasonal variations in external conditions. The goal is to maximize the utilization of free cooling, leveraging its reliance on outdoor temperatures and its minimal energy consumption profile. To achieve this, the Interquartile Range (IQR) is set to its widest extent, allowing for flexibility in operational thresholds. Similarly, percentiles are fixed at 100, ensuring that free cooling operates under all external temperature conditions.

The analysis of free cooling consumption during winter highlights the recommendation system's effective management of energy usage based on external temperatures, distinguishing between weekends and weekdays (Fig. 18). On weekends, free cooling activates below 10 °C, maintaining consumption predominantly between 0 % and 40 %, occasionally exceeding but generally adhering to the upper bound of 31 %. This approach signifies proactive energy conservation during quieter periods, optimizing usage within a narrow intensity range of 3–31 %. Conversely, weekdays see free cooling engagement below 16 °C,

Table 14
Percentile thresholds in hourly space heating consumption.

Season	Weekend Percentile	Weekday Percentile
Winter	97.5	100
Spring	97.5	100
Summer	97.5	100
Autumn	97.5	100

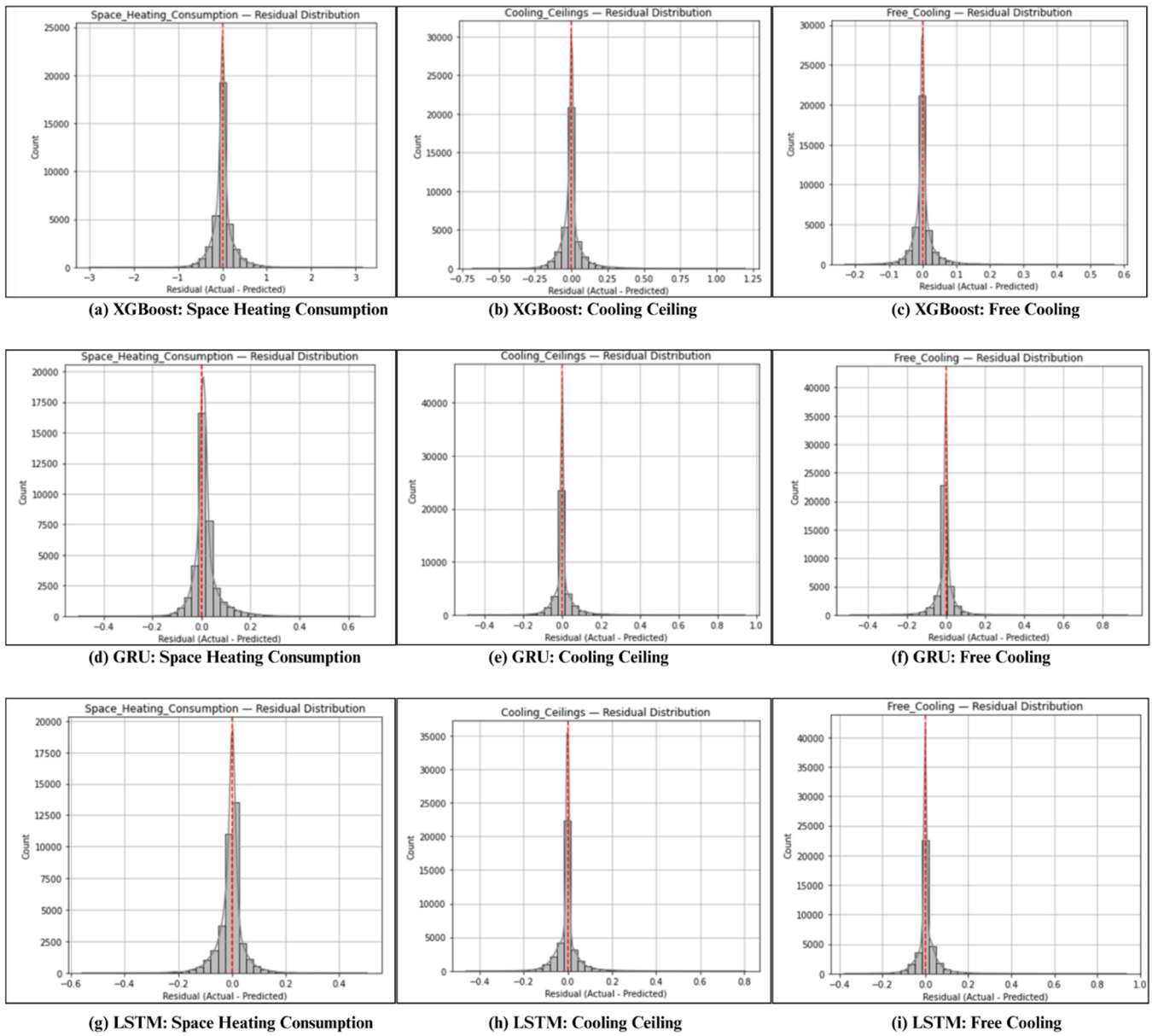


Fig. 13. Residual distributions of predicted vs. actual values (7 Days).

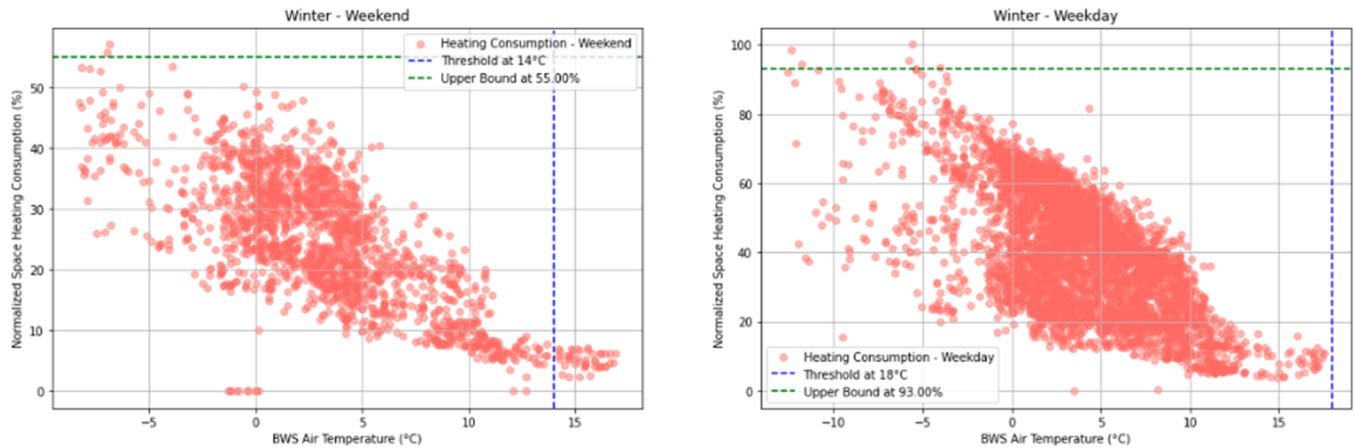


Fig. 14. Normalized space heating consumption vs external temperature (winter).

Table 15

Percentile thresholds in hourly cooling ceiling consumption.

Season	Weekend Percentile	Weekday Percentile
Winter	2.5	0
Spring	2.5	0
Summer	2.5	0
Autumn	2.5	0

managing consumption between 2.5 % and 36 %, with denser data clustering around 20–30 %. Despite a broader distribution, the system effectively controls energy expenditure within the upper limit of 36 %, ensuring efficient management during busier operational times.

During spring (Fig. 19), on weekends, free cooling consumption appears to peak around 17 °C BWS air temperature, with a maximum value of 49 %. Beyond this threshold, consumption declines significantly, indicating an optimal temperature range for effective free cooling. Similarly, for weekdays, shows a peak in free cooling consumption at around 19 °C, with more dispersed data points and a maximum value reaching 100 %. This suggests a higher utilization and variability of free cooling during weekdays compared to weekends

During autumn on weekends, Fig. 20 indicates that free cooling consumption peaks at around 15 °C, with a maximum value of 54 %. This pattern is consistent with the autumn data, showing a decrease in consumption as temperatures rise beyond the threshold. On weekdays, the threshold temperature for maximum free cooling consumption is 18

°C, similar to autumn weekdays, but with a lower maximum value of 80 %. The data again highlights greater variability and usage of free cooling on weekdays compared to weekends, with spring showing slightly different optimal temperature ranges compared to autumn. Figs. 21 and 22

Table 17 summarizes how the graphical interpretation of energy consumption patterns informs the recommendations for each system and season.

6. Discussion

The analysis of the proposed recommendation system reveals its significant effectiveness in managing both heating and cooling requirements across different seasons and occupancy patterns. During winter, the system demonstrates a robust capability to adjust heating operations based on specific temperature thresholds and consumption limits. This adaptability ensures that heating is utilized conservatively during weekends when occupancy is typically lower, while scaling up efficiently on weekdays to meet higher demand. Such strategic management not only optimizes energy consumption but also enhances operational efficiency by aligning heating usage closely with actual occupancy needs. Graphical interpretation using GPT-4 has enabled precise identification of these thresholds and patterns, providing actionable insights that refine the system's operational strategy.

Similarly, in summer, the recommendation system excels in

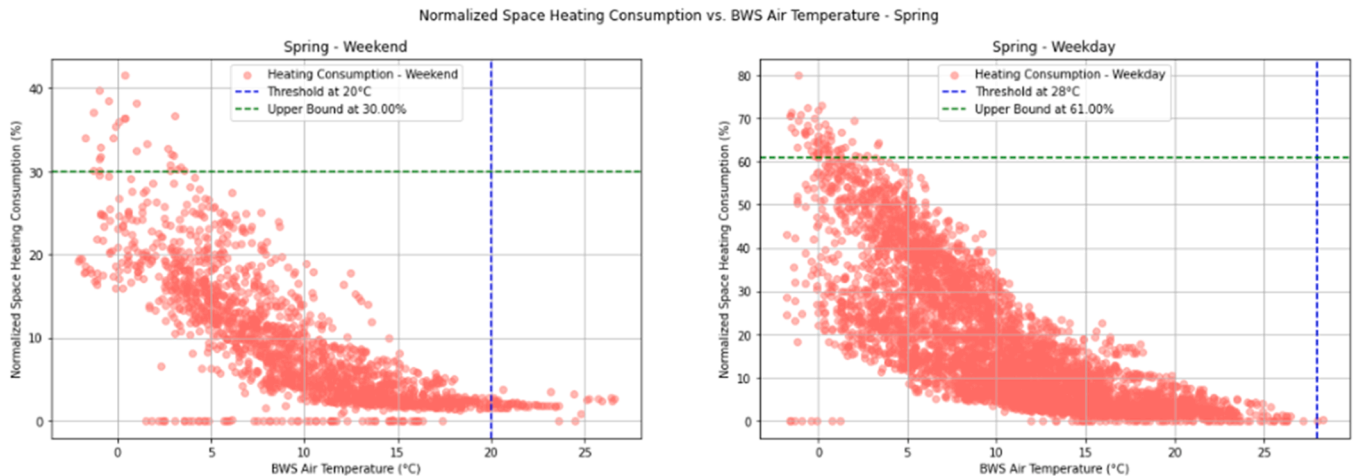


Fig. 15. Normalized space heating consumption vs external temperature (spring).

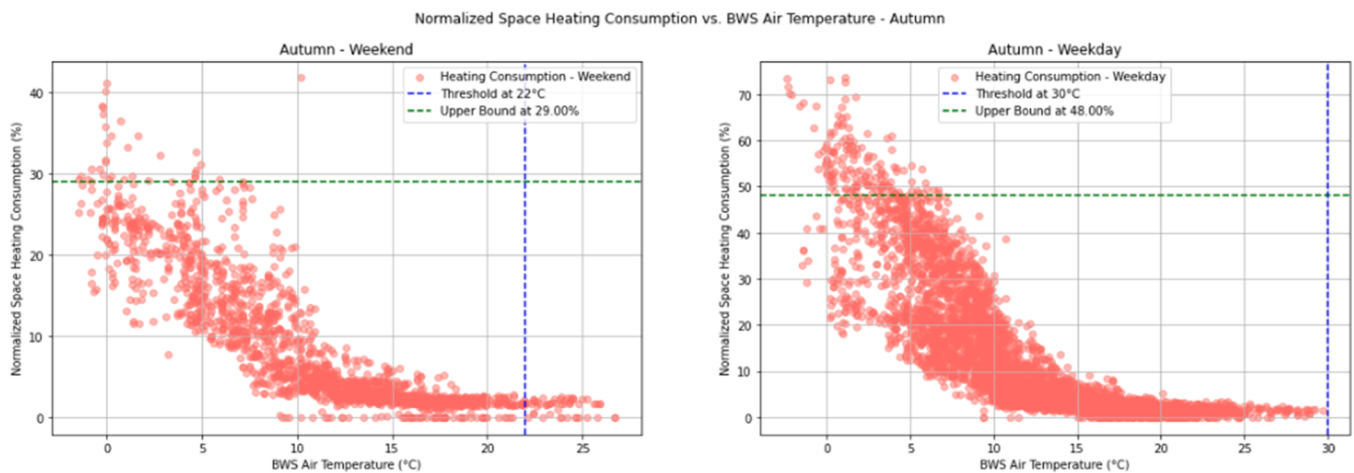


Fig. 16. Normalized space heating consumption vs external temperature (autumn).

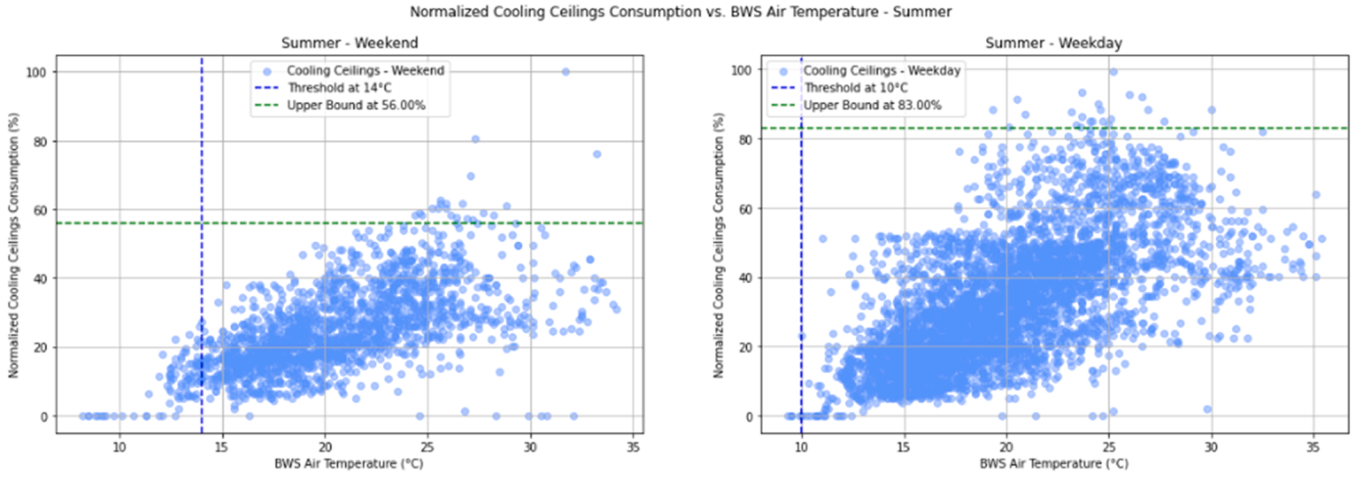


Fig. 17. Normalized cooling ceiling vs external temperature (summer).

Table 16

Percentile thresholds in hourly free cooling consumption.

Season	Weekend Percentile	Weekday Percentile
Winter	100	100
Spring	100	100
Summer	100	100
Autumn	100	100

managing space cooling demands by setting strict upper bounds and temperature thresholds. This approach ensures that cooling is engaged judiciously, minimizing energy consumption while maintaining comfortable indoor conditions. By deploying cooling operations only when temperatures exceed predefined thresholds, the system effectively balances energy efficiency with comfort requirements. The GPT-4 model's analysis of seasonal cooling patterns has highlighted optimal trigger temperatures and intensity bounds, demonstrating how precise graphical insights translate into reduced operational costs and enhanced building management efficiency. This strategic deployment not only reduces operational costs but also supports sustainable building practices, emphasizing the system's role in achieving optimal energy use and enhancing overall building management efficiency.

Furthermore, the system's handling of free cooling during both winter and summer highlights its adaptive capability in response to varying external conditions. By regulating free cooling within defined intensity bounds and temperature limits tailored to weekend and

weekday dynamics, the system optimizes energy efficiency without compromising comfort. The integration of GPT-4 in interpreting free cooling patterns has underscored its effectiveness in maximizing energy savings while maintaining comfort, reinforcing its role in sustainable energy practices and operational efficiency across different seasons. This approach underscores its effectiveness in promoting sustainable energy practices and improving operational efficiency across different seasons, contributing significantly to overall energy savings and building performance.

Even though ceiling cooling does not represent the full cooling load, it not only represents the biggest share of consumption but also the most important part allocated to comfort cooling. This underscores the critical role of ceiling cooling in maintaining indoor climate control and enhancing occupant comfort, highlighting the importance of optimizing this component within the overall energy management strategy. The GPT-4 model's insights into ceiling cooling patterns have further emphasized the need for precise adjustments to ensure optimal comfort and efficiency.

6.1. Applications of the proposed recommendation system

The proposed recommendation system offers versatile solutions tailored to optimize heating consumption, cooling ceilings, and free cooling strategies. Here, we explore three distinct applications where this innovative system can deliver substantial benefits across various sectors and operational contexts.

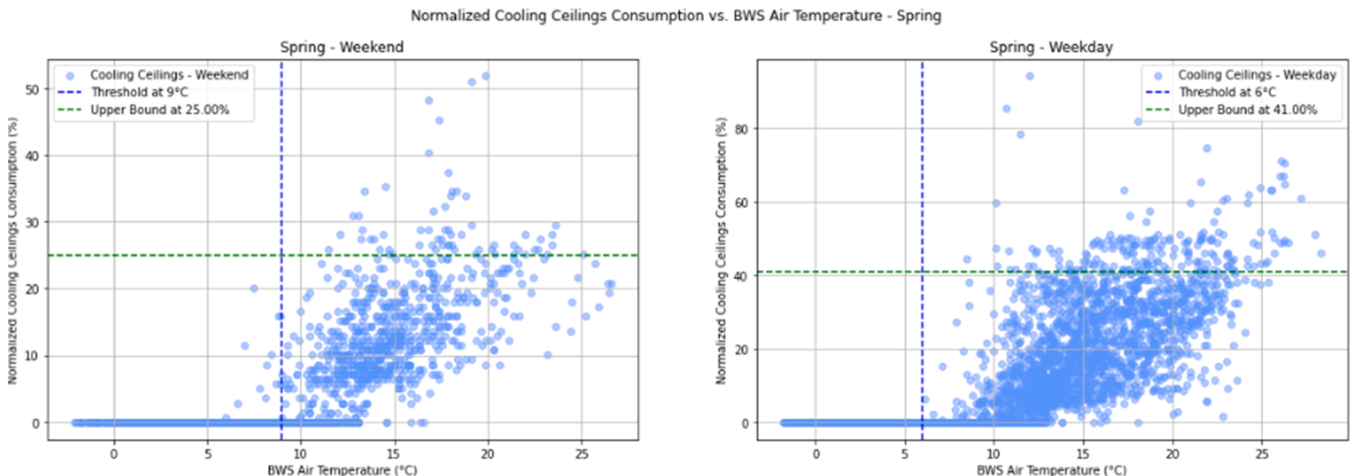


Fig. 18. Normalized cooling ceiling vs external temperature (spring).

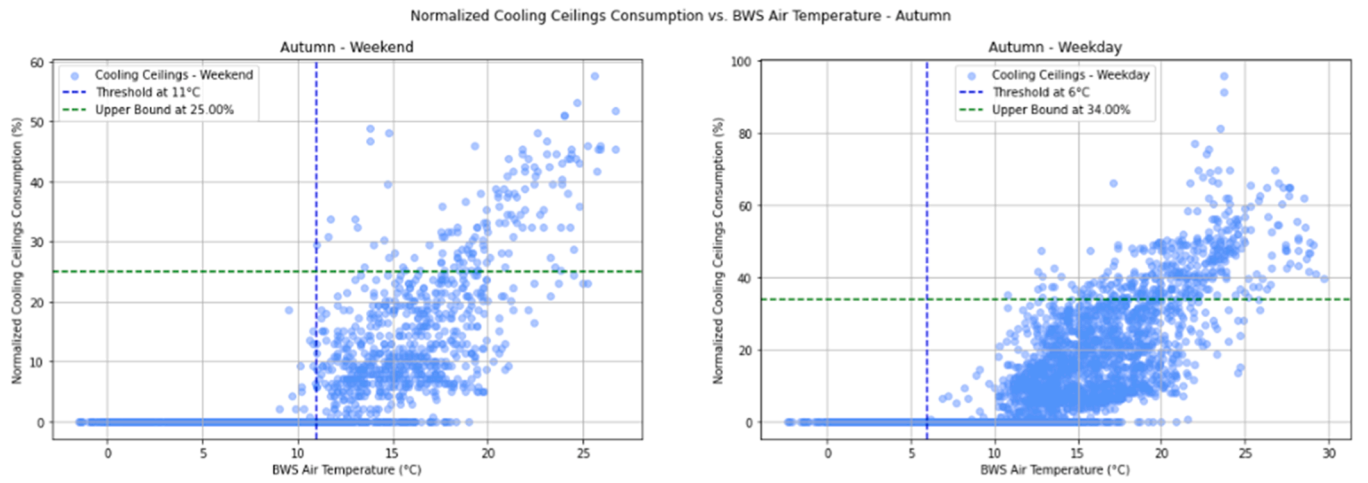


Fig. 19. Normalized cooling ceiling vs external temperature (autumn).

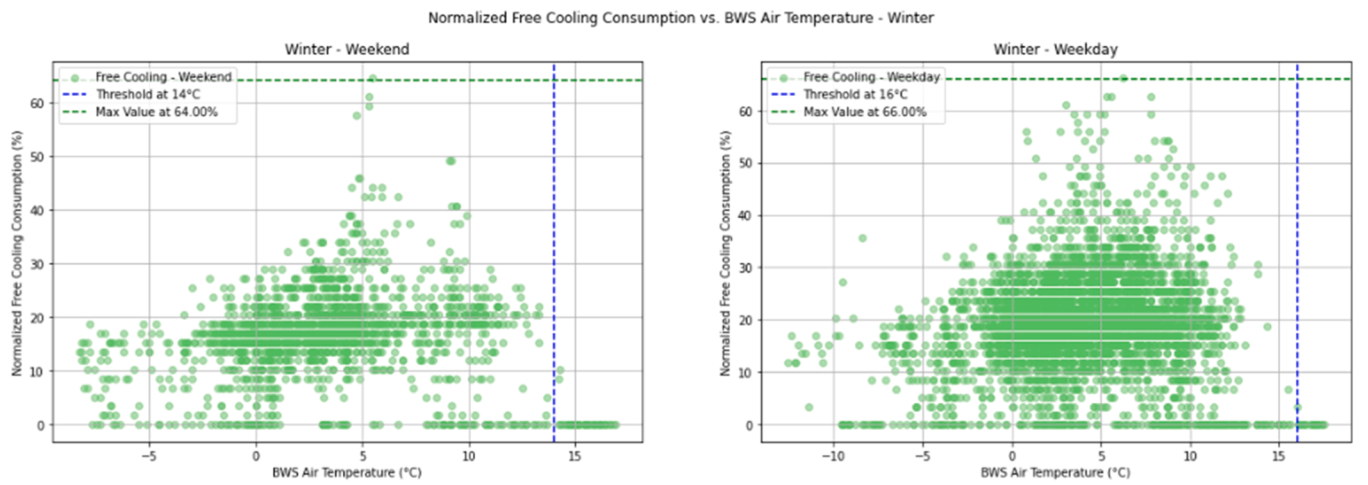


Fig. 20. Normalized free cooling vs external temperature (winter).

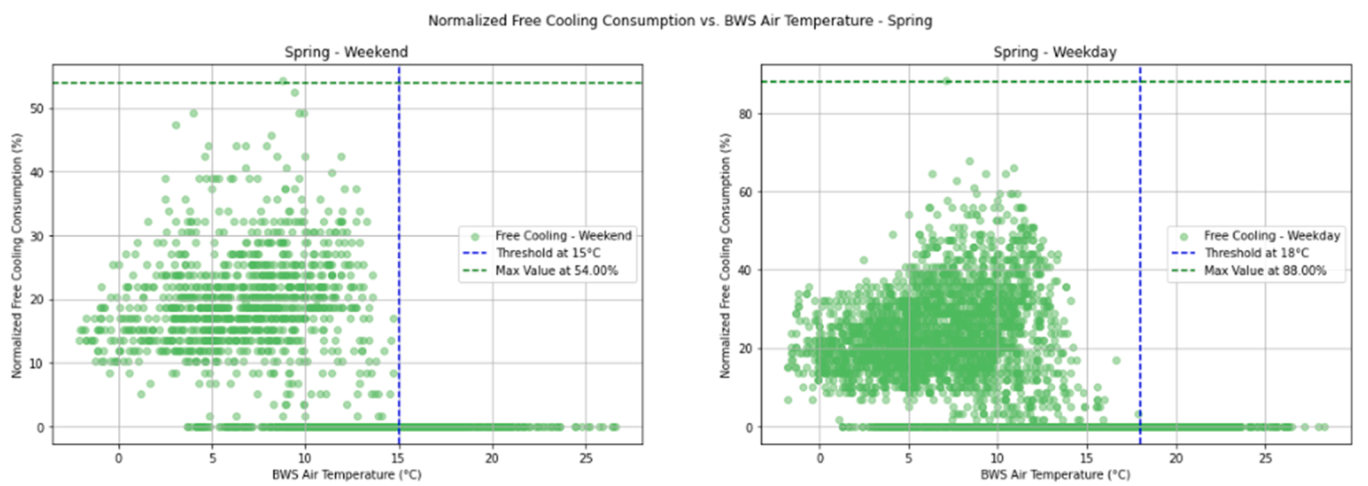


Fig. 21. Normalized free cooling vs external temperature (spring).

i. Building Management Companies

This recommendation system would be useful for building management companies where they can use it for the management and efficient

operation of their properties. Incorporating the system into the building's automation systems they can control the operation of heating, cooling ceilings and free cooling according to the current environmental conditions and the occupancy levels [32]. This proactive approach

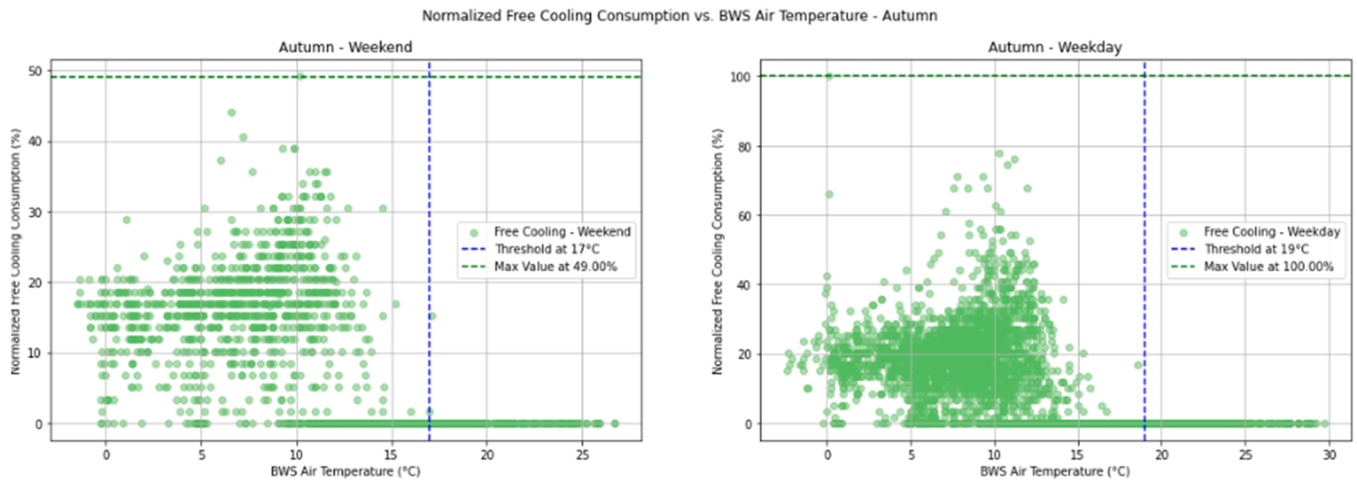


Fig. 22. Normalized free cooling vs external temperature (autumn).

ensures that occupants of the building are comfortable by providing them with necessary conditioned air while at the same time avoiding wastage of energy during off-peak hours. This flexibility also makes the system efficient in performance throughout the year and saves operational costs as well as the carbon footprints. Furthermore, through the system, detailed analytics are obtained to allow the building managers to plan for the energy demands and implement the necessary improvements to support sustainable building operational services and tenants' satisfaction.

ii. Facility Managers in Commercial Buildings

The recommendation system for heating consumption, cooling ceilings, and free cooling have the potential of bringing much benefits to the facility managers of the commercial buildings. It will help them to centralize the HVAC systems and control energy consumption patterns based on weather changes and occupancy rates. At times of high heating or cooling need, the system's predictive ability assists in anticipating climate conditions in order to regulate settings to reduce energy consumption while meeting a set required temperature [24]. Such management also helps to minimize utility costs and also helps the lifespan of HVAC equipment because they are not strained during times of extreme temperatures. In addition, the features of real-time monitoring & reporting help the facility managers to analyze the performance indicators, detect the possible problems, and apply the corrective actions to improve the energy management more effectively and efficiently.

iii. Energy Consultants and Sustainability Experts

The recommendation system can also be used by energy consultants and sustainability specialists to offer specific recommendations to clients who are interested in improving the energy efficiency of buildings and meeting sustainability objectives. Through comprehensive energy surveys and incorporating features of the system that allows for prediction, consultants are able to devise specific approaches to heating and cooling as well as free cooling depending on the characteristics of the building and its usage [33]. This approach makes it easy to adopt energy efficient practices that meet the current legal and industrial standards thus lowering the operational expenses as well as emissions of greenhouse gases. Further, continuous analysis and optimization made possible by the system provide consultants a way to substantiate and prove energy savings, and continue to recommend adjustments, thus ensuring lasting positive change for their clients' business and the environment.

6.2. Limitations

While the proposed LSTM architecture demonstrates strong predictive performance for the studied building, its generalization to different building types remains a challenge. Variations in thermal behavior, stemming from differences in building envelope characteristics, HVAC configurations, and occupancy schedules, can significantly influence energy consumption patterns. Models trained on a single building may fail to account for these unique operational profiles when applied elsewhere. Additionally, diversity in control systems poses a limitation, as differences in control logic, setpoints, and operational strategies may reduce the transferability of learned patterns. A further constraint lies in data availability; many buildings lack the high-resolution historical datasets necessary for effective LSTM training, limiting the feasibility of deploying such models in data-scarce environments.

7. Conclusion

This study presents a robust methodology for optimizing energy consumption in building environments through the development of a LSTM neural network model. The model effectively predicts the efficiency levels of space heating, ceiling cooling, and free cooling systems on both hourly and daily timescales, enabling precise and adaptive energy management. Key findings demonstrate that the LSTM model achieves high predictive accuracy, with R-squared values indicating its ability to explain up to 97 % of the variance in energy consumption across different scenarios. These results validate the model's robustness and effectiveness in forecasting energy needs, thereby supporting efficient operational planning. To assess the relative performance of the LSTM model, it is systematically compared with two benchmark models, Gated Recurrent Unit (GRU) and XGBoost, across multiple timescales. The results consistently show that LSTM outperforms GRU and XGBoost in predictive accuracy, particularly for long-range forecasts, highlighting its suitability for building energy management applications. Moreover, the integration of real-time weather data from a specialized weather station atop the ECB skyscraper enhances the model's relevance and accuracy. By examining historical consumption patterns and meteorological data, the system provides actionable insights for building managers, facilitating informed decision-making to optimize HVAC operations. The interpretation of graphical data using GPT-4 has further enriched the analysis, offering detailed insights into energy consumption patterns and system performance. This capability allows for nuanced adjustments and a deeper understanding of how different variables impact energy efficiency. The proposed recommendation system excels in adapting to varying seasonal and occupancy conditions, as evidenced by its strategic management of heating and cooling demands.

Table 17
Seasonal energy management recommendations based on graphical interpretation.

Season	System	Recommendation	Interpretation - Weekends	Interpretation - Weekdays
Winter	Space Heating (SH)	Implement automated, adaptive control algorithms.	Minimal operation at temperatures <14 °C.	More intense operation even when external temperatures rise up to 18 °C due to higher occupancy.
	Cooling Ceilings (CC)	Inactive during winter.	No intensity above 0 % is recommended.	No intensity above 0 % is recommended.
	Free Cooling (FC)	Maximize use during mild temperatures.	Operate at full capacity when active, reducing energy needed for heating.	Operate at full capacity when active, reducing energy needed for heating.
Summer	Space Heating (SH)	Optimize energy efficiency.	Intensity should be 0–4 % when temperatures <29 °C.	Intensity should be 0–6.0 % when temperatures <35 °C.
	Cooling Ceilings (CC)	Manage cooling ceilings based on temperature.	Should not operate when temperatures >14 °C, with intensity between 0–56 %.	Should not operate when temperatures >10 °C, with intensity between 0–83 %.
	Free Cooling (FC)	Maximize the use of Free Cooling system.	Use to replace heating/cooling demands during mild temperatures.	Use to replace heating/cooling demands during mild temperatures.
Spring	Space Heating (SH)	Adjust operational settings.	Activate below 20 °C, limiting intensity to 0–30 %.	Activate below 28 °C, limiting intensity to 0–61 %.
	Cooling Ceilings (CC)	Adjust system operation.	Inactive when temperatures >9 °C, with intensity between 0–25 %.	Inactive when temperatures >6 °C, with intensity between 0–41 %.
	Free Cooling (FC)	Efficiently utilize Free Cooling.	Maximize intensity within the cooling range.	Maximize intensity within the cooling range.
Autumn	Space Heating (SH)	Optimize heating intensity.	Reduce intensity to a maximum of 29 % when temperatures drop below 22 °C.	Reduce intensity to a maximum of 48 % when temperatures drop below 30 °C.
	Cooling Ceilings (CC)	Reduce cooling activity.	Inactive when temperatures >11 °C, with intensity between 0–25 %.	Inactive when temperatures >6 °C, with intensity between 0–34 %.
	Free Cooling (FC)	Maximize Free Cooling usage.	Set intensity to maximum whenever needed.	Set intensity to maximum whenever needed.

In winter, the system adjusts heating operations based on specific temperature thresholds, optimizing energy use during both weekends and weekdays. Similarly, in summer, the system judiciously manages cooling activities, ensuring energy-efficient operation while maintaining indoor comfort. The handling of free cooling further underscores the system's adaptive capabilities. By regulating free cooling based on defined intensity bounds and temperature limits tailored to weekend and weekday dynamics, the system maximizes energy efficiency without

compromising comfort.

CRedit authorship contribution statement

Fernando Almeida: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mauro Castelli:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Nadine Côte-Real:** Writing – review & editing, Validation, Supervision. **Camilla Fallarino:** Methodology, Investigation, Data curation, Conceptualization. **Luca Manzoni:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Fernando Almeida reports article publishing charges was provided by NOVA University Lisbon NOVA Information Management School. Fernando Almeida reports a relationship with European Central Bank that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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